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Lin et al.

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(54) **INTEGRATED CIRCUIT STRUCTURE WITH FILLED RECESSES**

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(51) **Int. Cl.**

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H01L 23/00	(2006.01)
H01L 23/31	(2006.01)
H01L 23/522	(2006.01)
H01L 25/00	(2006.01)
H01L 25/065	(2023.01)

(52) **U.S. Cl.**

CPC ***H01L 23/53295*** (2013.01); ***H01L 23/3128*** (2013.01); ***H01L 23/5226*** (2013.01); ***H01L 24/09*** (2013.01); ***H01L 24/17*** (2013.01); ***H01L 25/0655*** (2013.01); ***H01L 25/50*** (2013.01); *H01L 2224/0401* (2013.01)

(58) **Field of Classification Search**

CPC H01L 23/53295; H01L 23/3128; H01L 23/5226; H01L 24/09; H01L 24/17; H01L 25/0655; H01L 25/50; H01L 2224/0401; H01L 257/69

USPC 257/690

See application file for complete search history.

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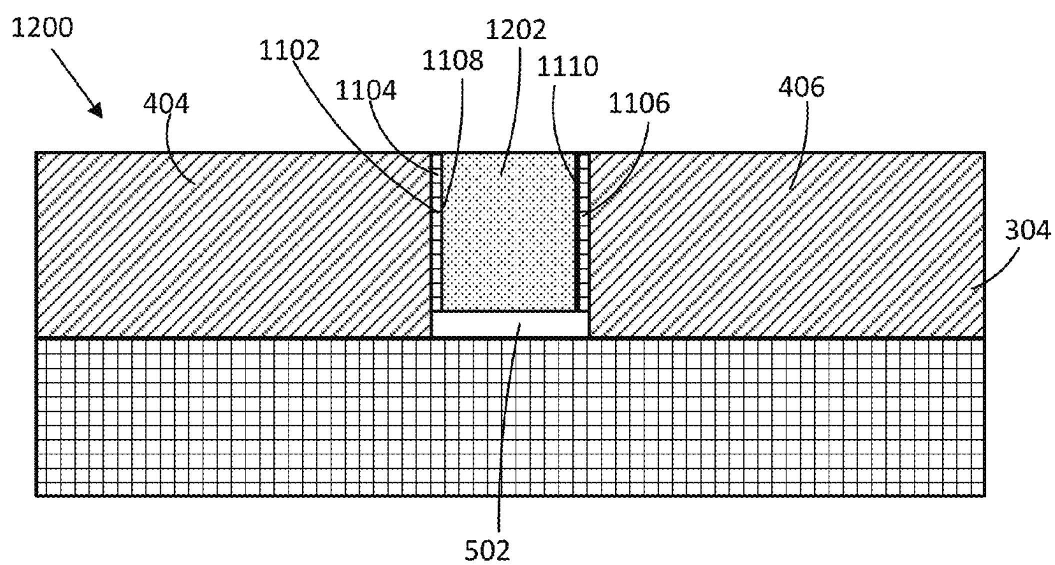
(74) *Attorney, Agent, or Firm* — Haley Guiliano LLP

(57) **ABSTRACT**

Disclosed herein are IC structures, packages, and devices that include recesses processed via selective growth. An example integrated circuit (IC) structure, includes a first dielectric material, a second dielectric material on the first dielectric material, and a recess in the second dielectric material, wherein the recess includes a bottom, a top, and sidewalls. The IC further includes a first material within the recess and at a bottom of the recess, wherein the first material includes a metal and oxygen, a self-assembled monolayer (SAM) material, or an organic material, and a second material within the recess and between the first material and the top of the recess, wherein the second material is in contact with the sidewalls of the recess.

20 Claims, 18 Drawing Sheets



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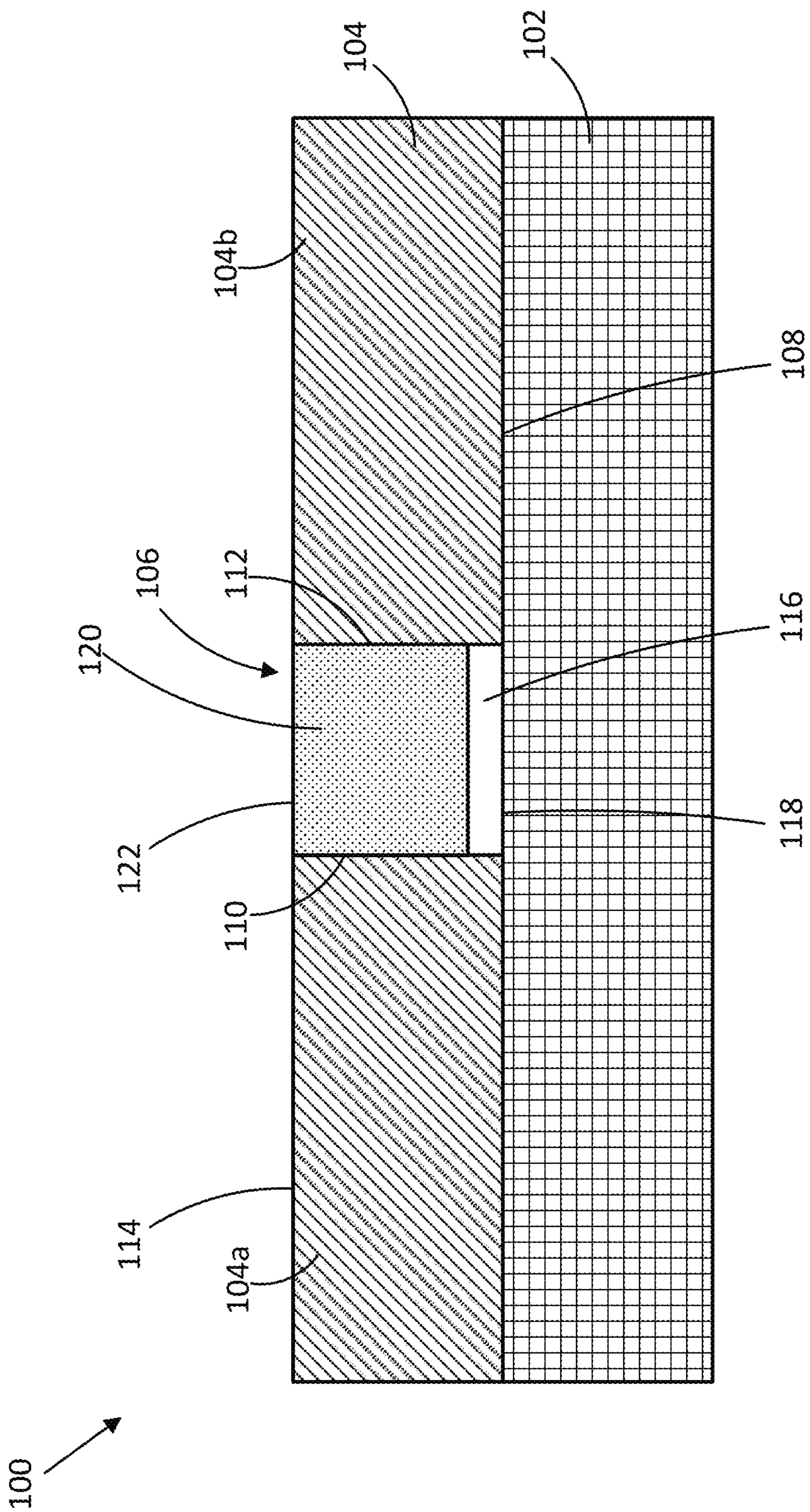


FIGURE 1

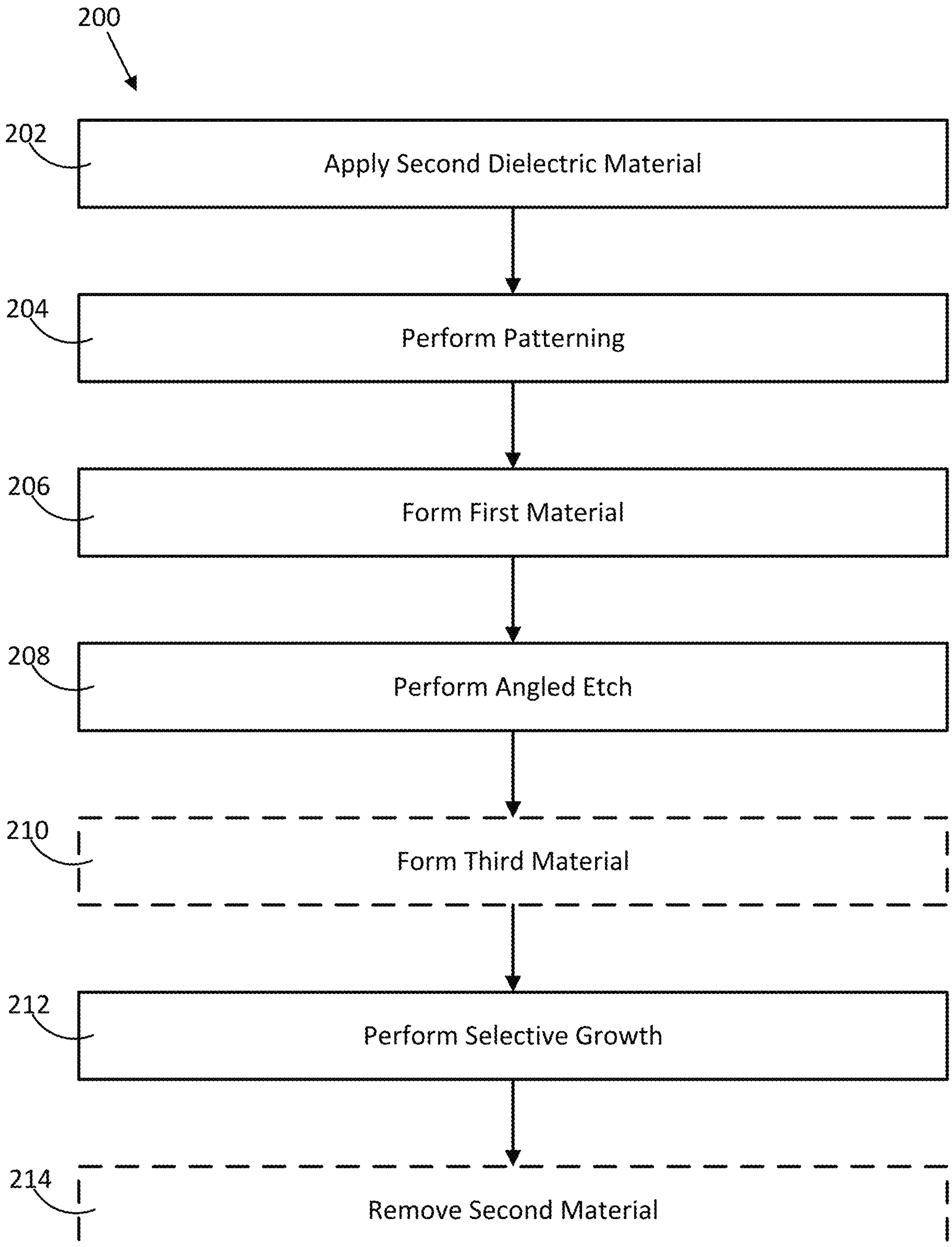

200
202
Apply Second Dielectric Material
204
Perform Patterning
206
Form First Material
208
Perform Angled Etch
210
Form Third Material
212
Perform Selective Growth
214
Remove Second Material

FIGURE 2

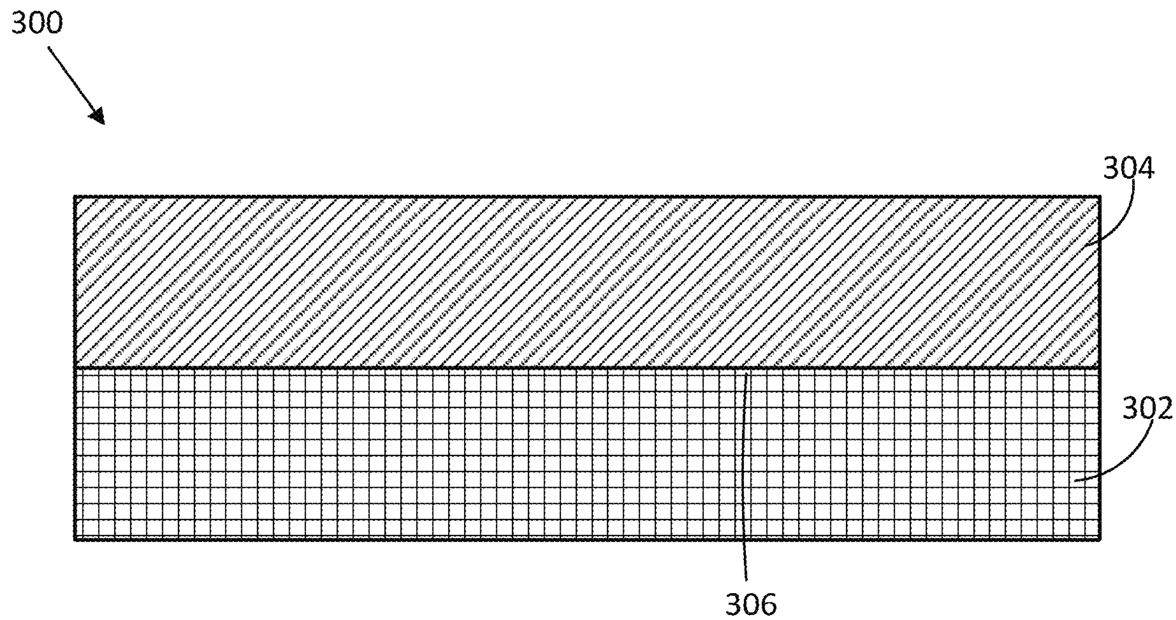


FIGURE 3

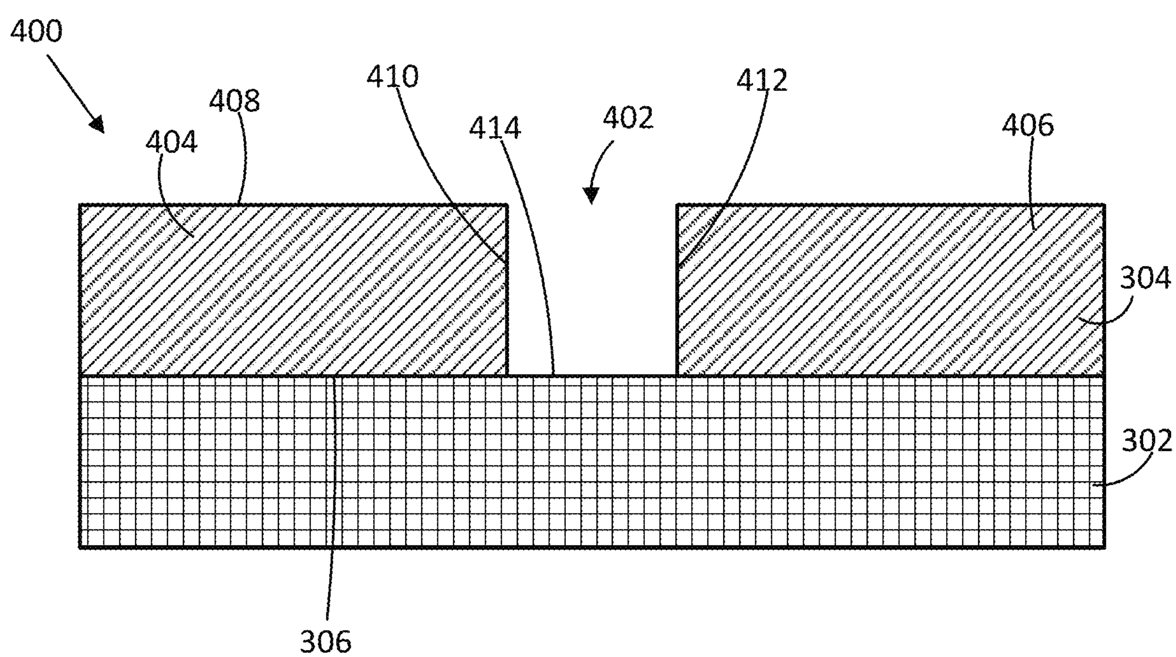


FIGURE 4

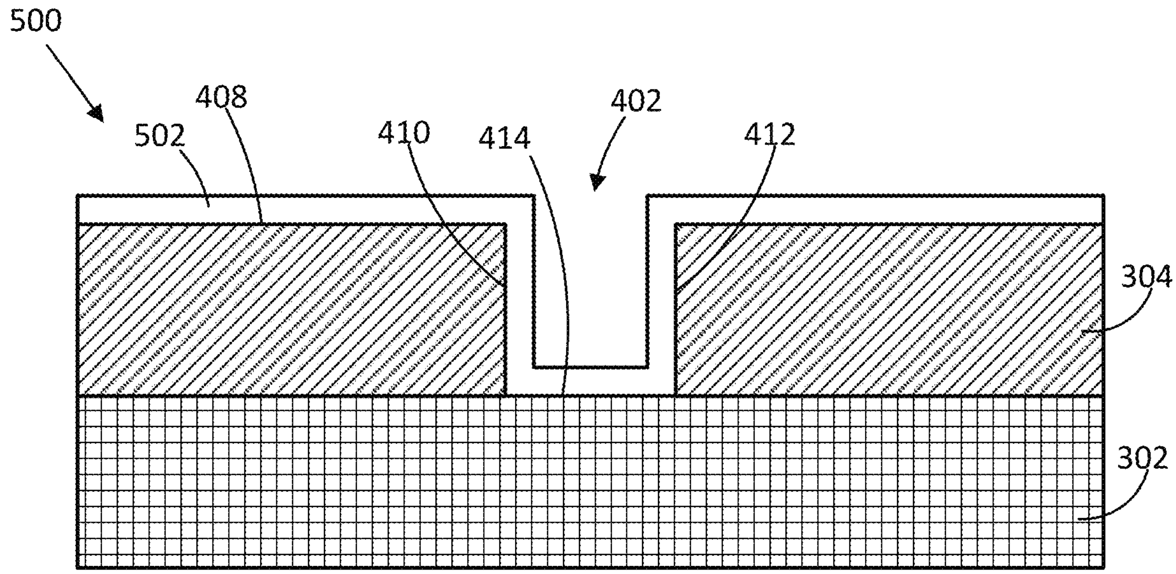


FIGURE 5

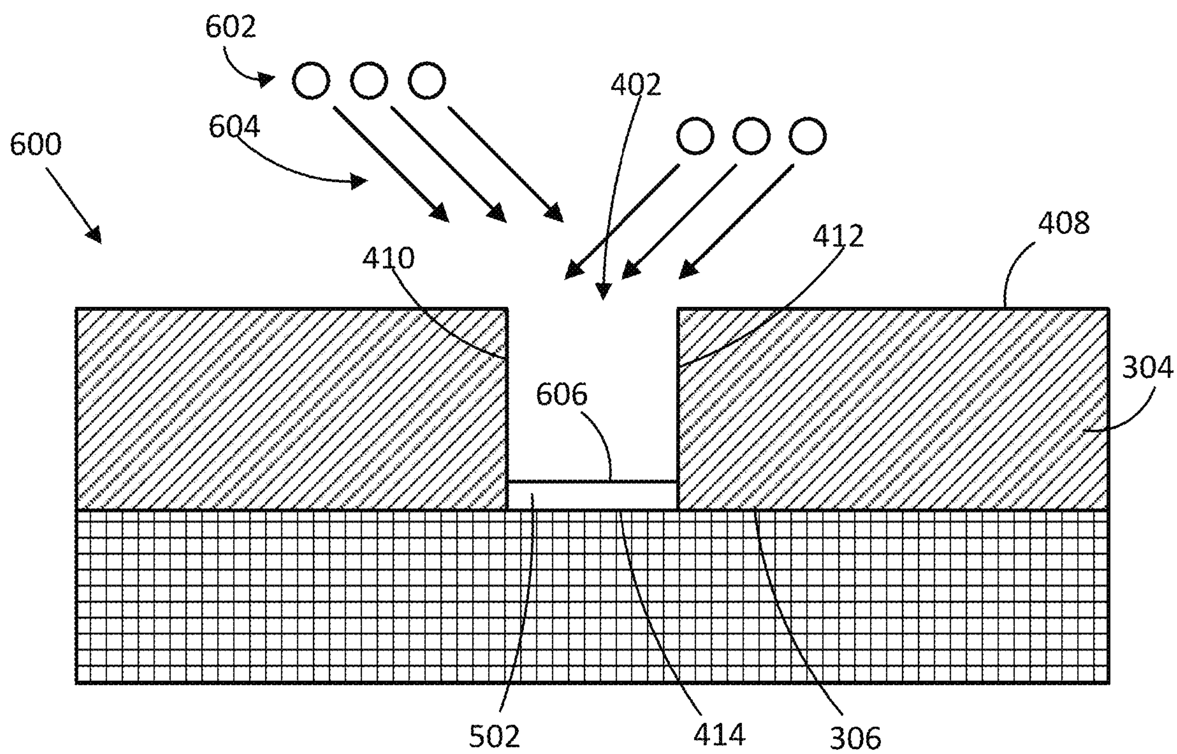


FIGURE 6

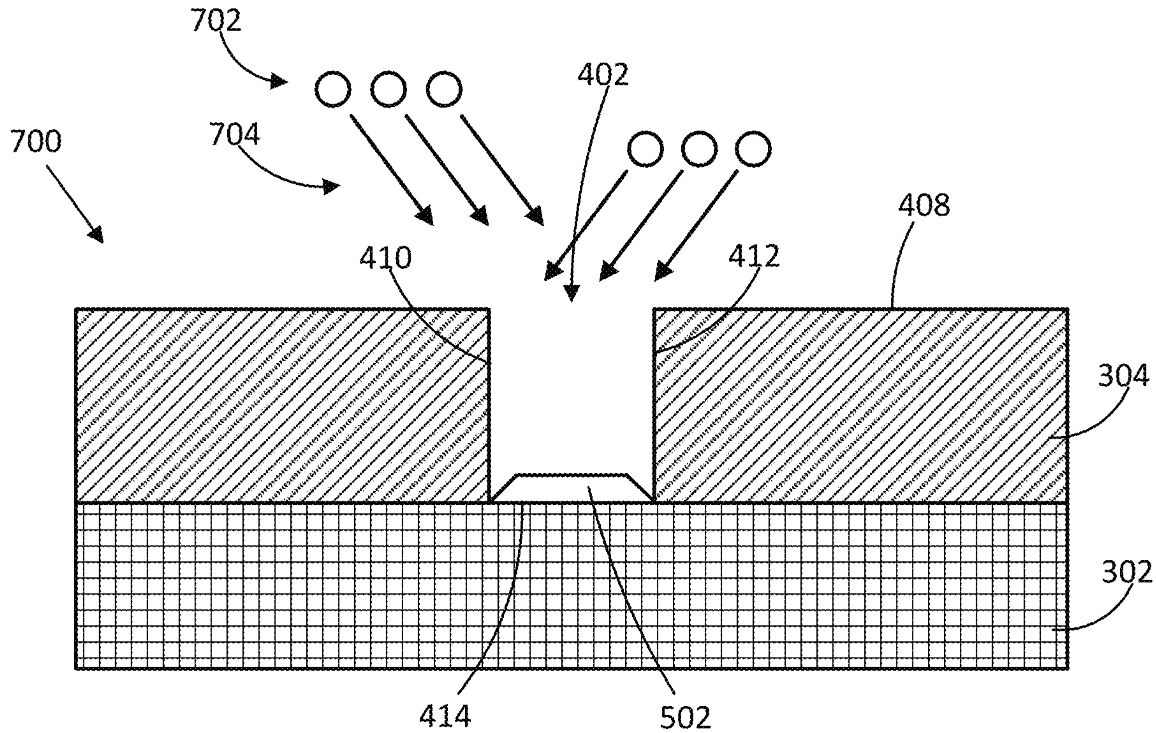


FIGURE 7

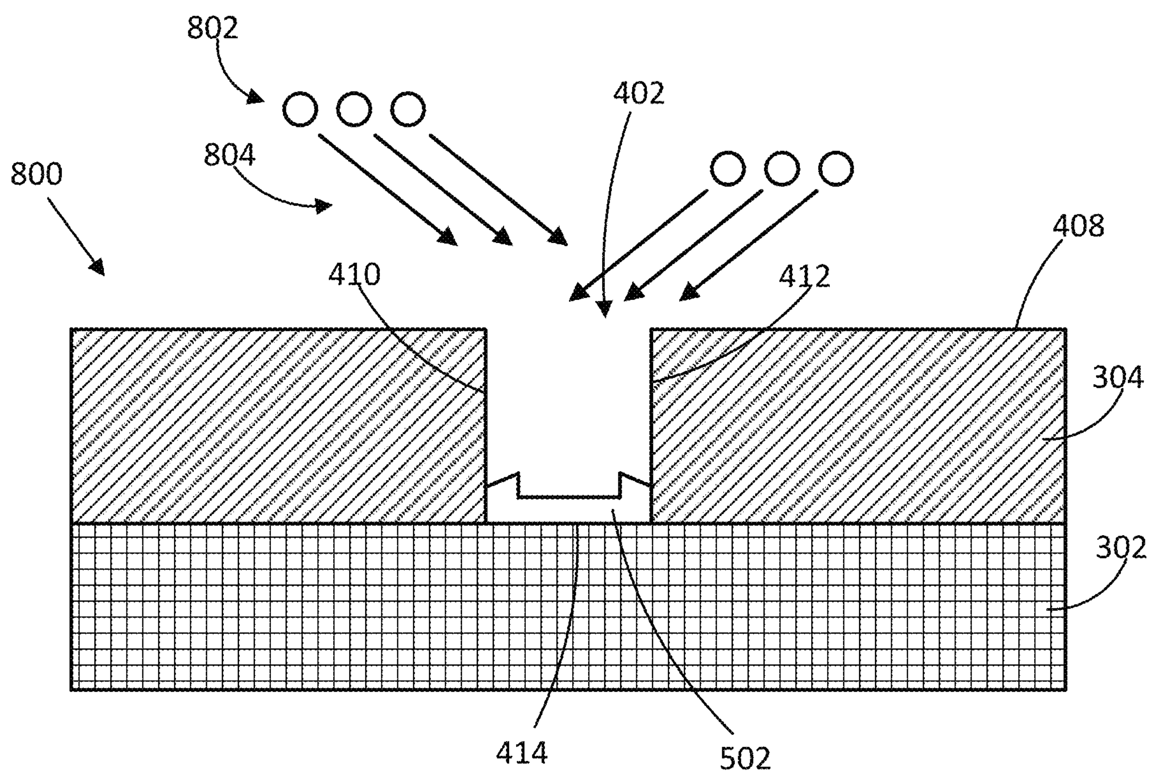


FIGURE 8

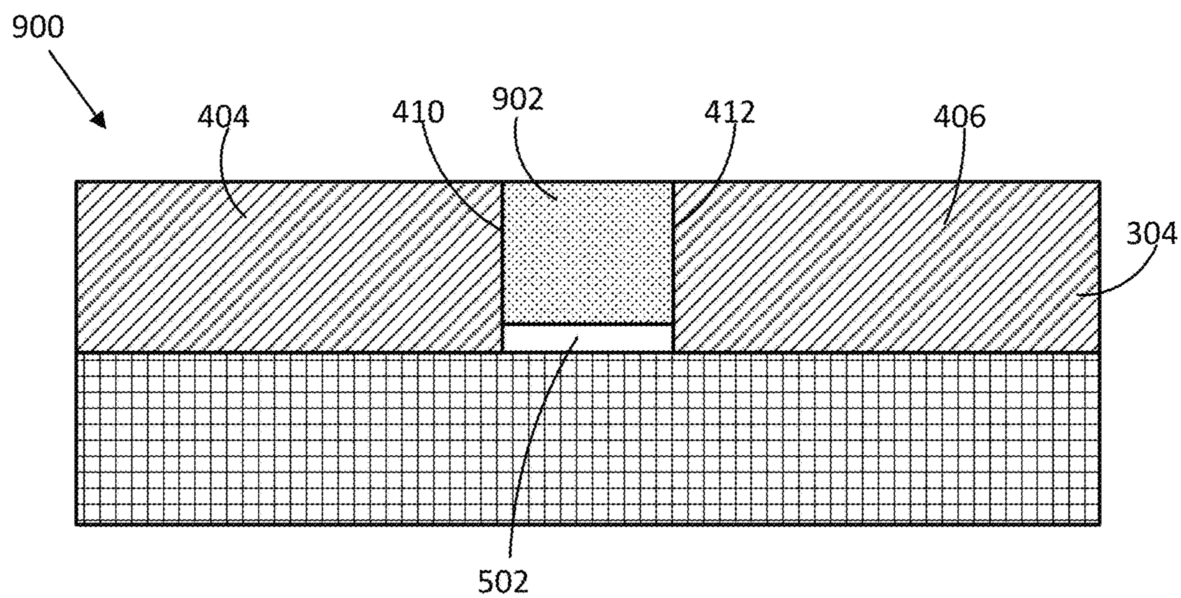


FIGURE 9

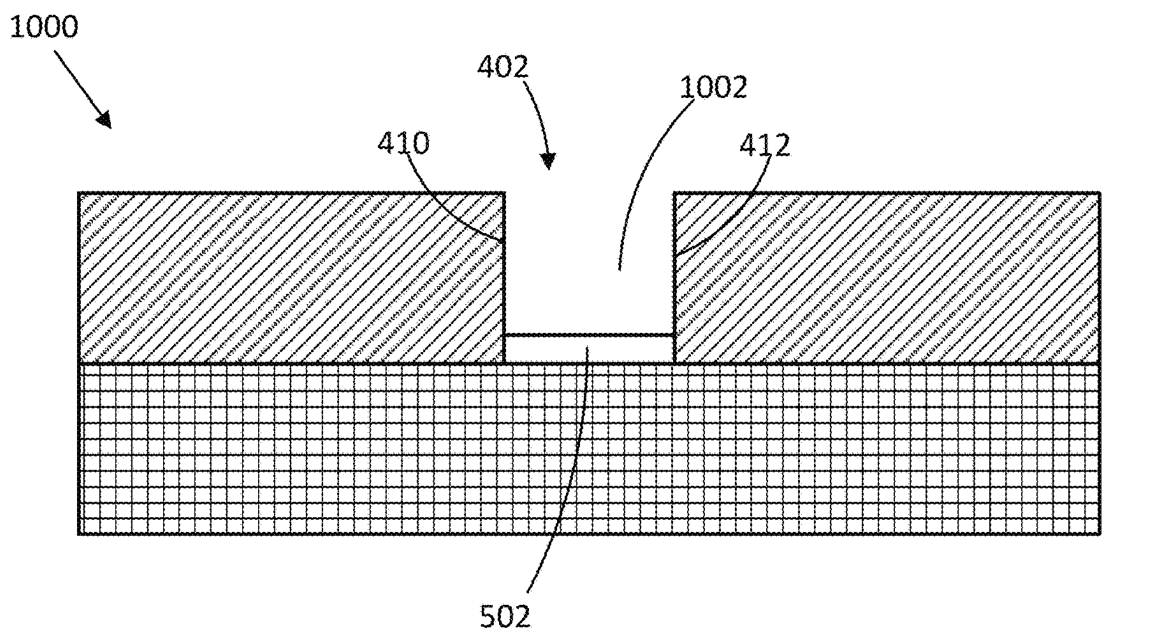


FIGURE 10

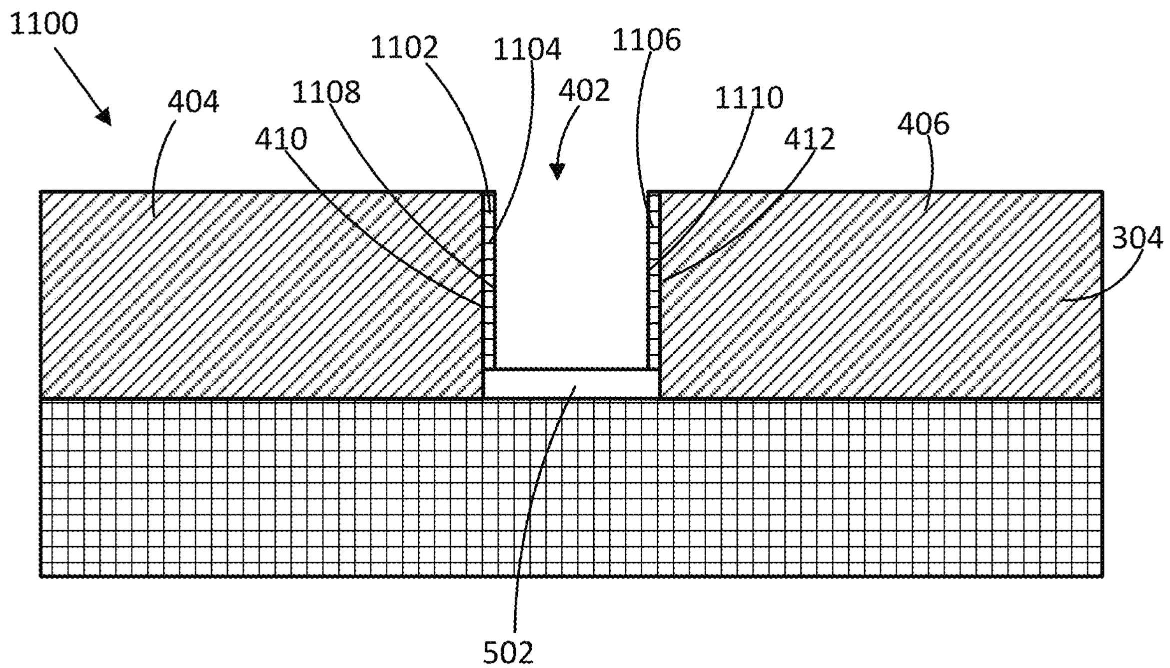


FIGURE 11

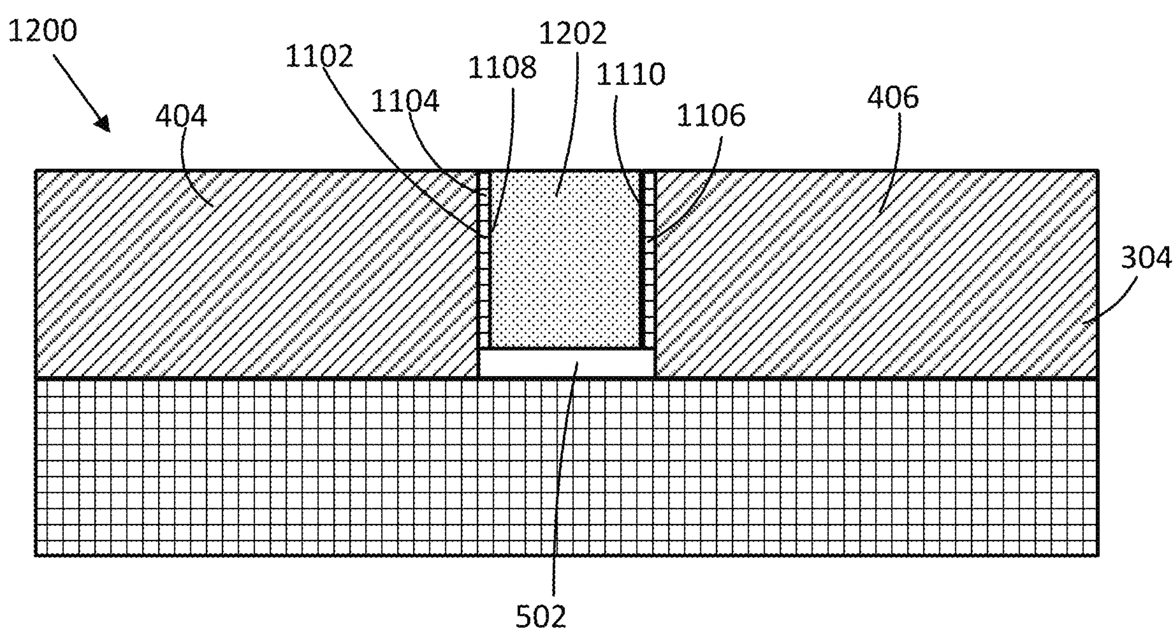


FIGURE 12

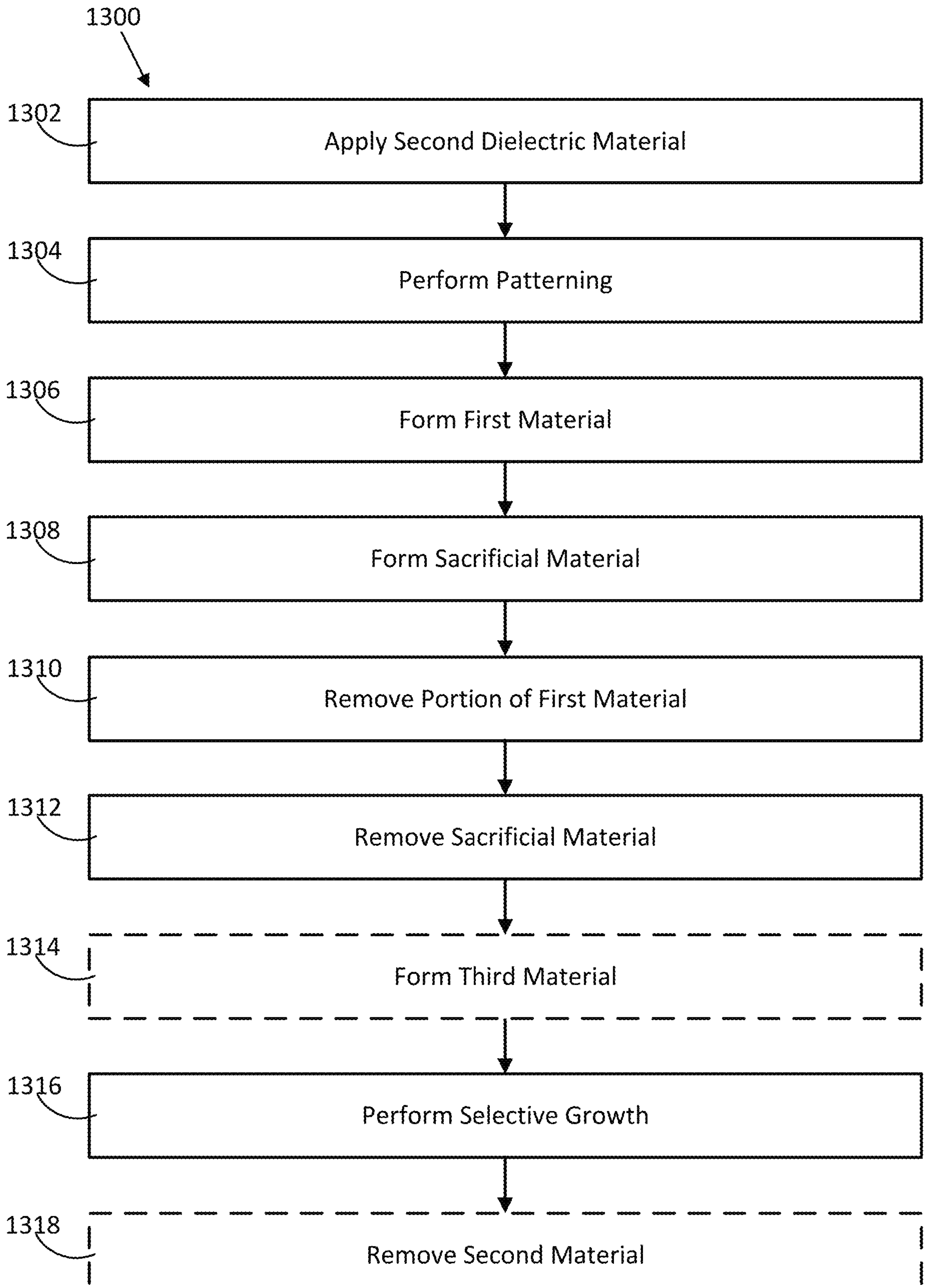

1300
1302
Apply Second Dielectric Material
1304
Perform Patterning
1306
Form First Material
1308
Form Sacrificial Material
1310
Remove Portion of First Material
1312
Remove Sacrificial Material
1314
Form Third Material
1316
Perform Selective Growth
1318
Remove Second Material

FIGURE 13

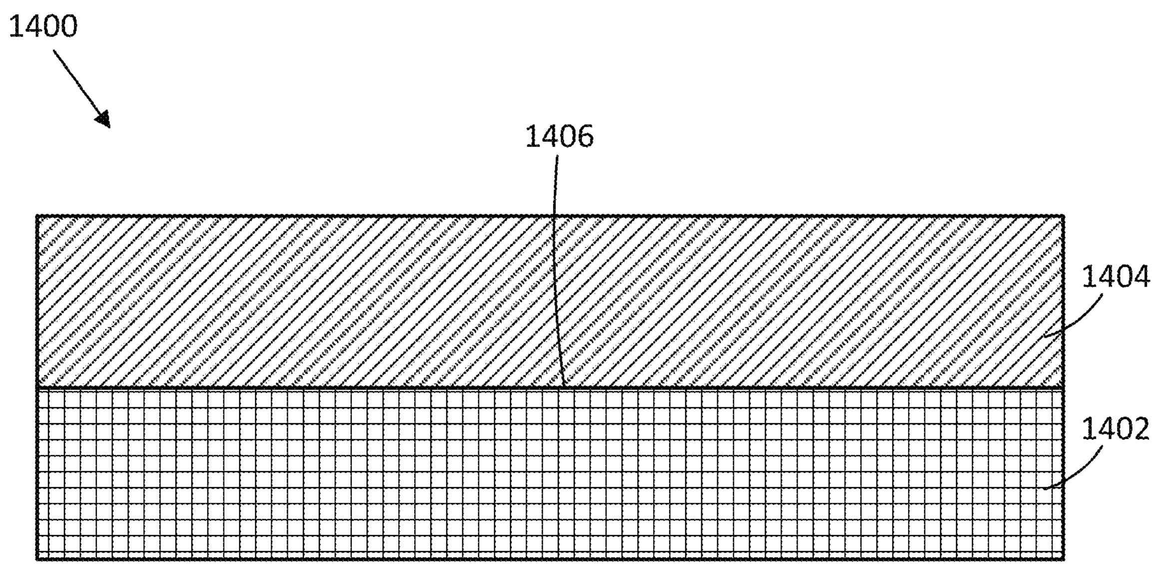


FIGURE 14

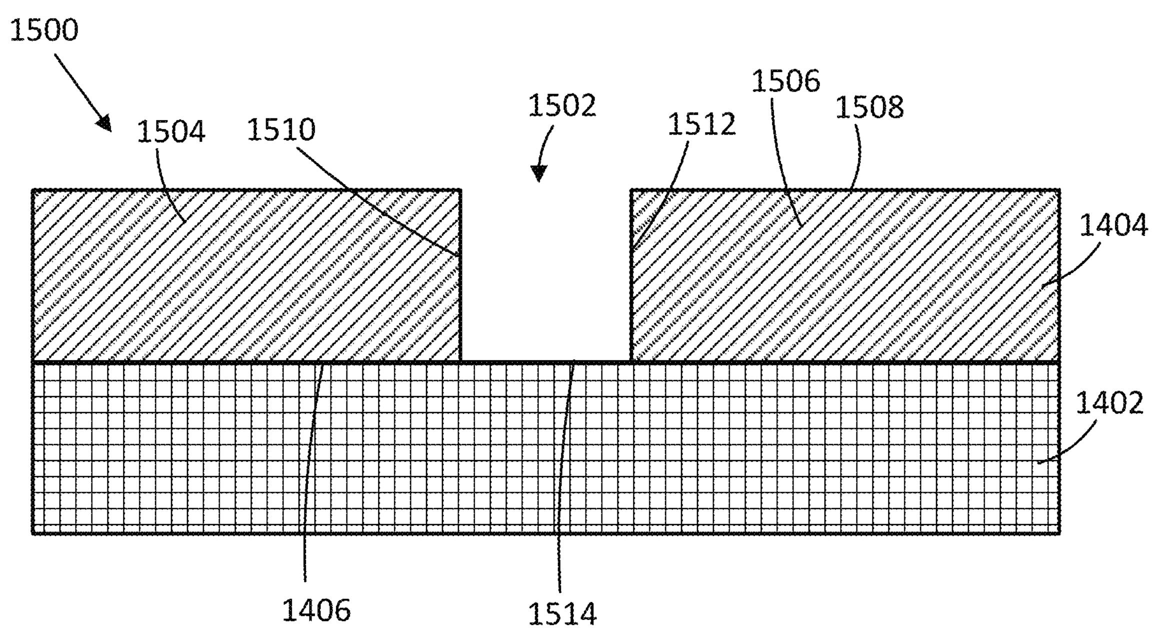


FIGURE 15

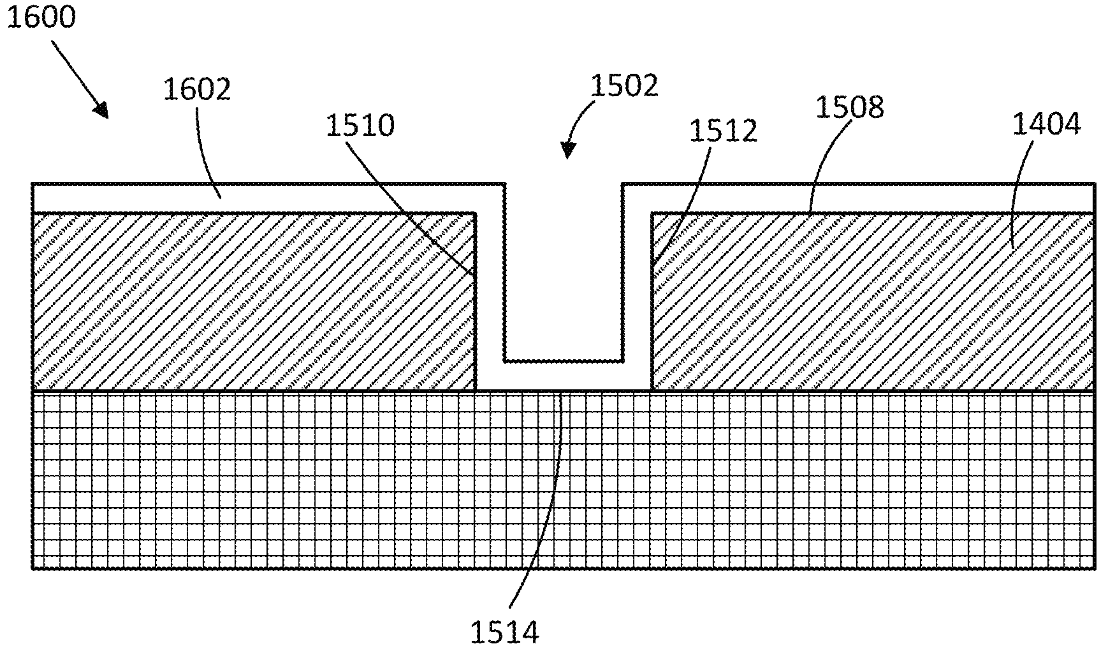


FIGURE 16

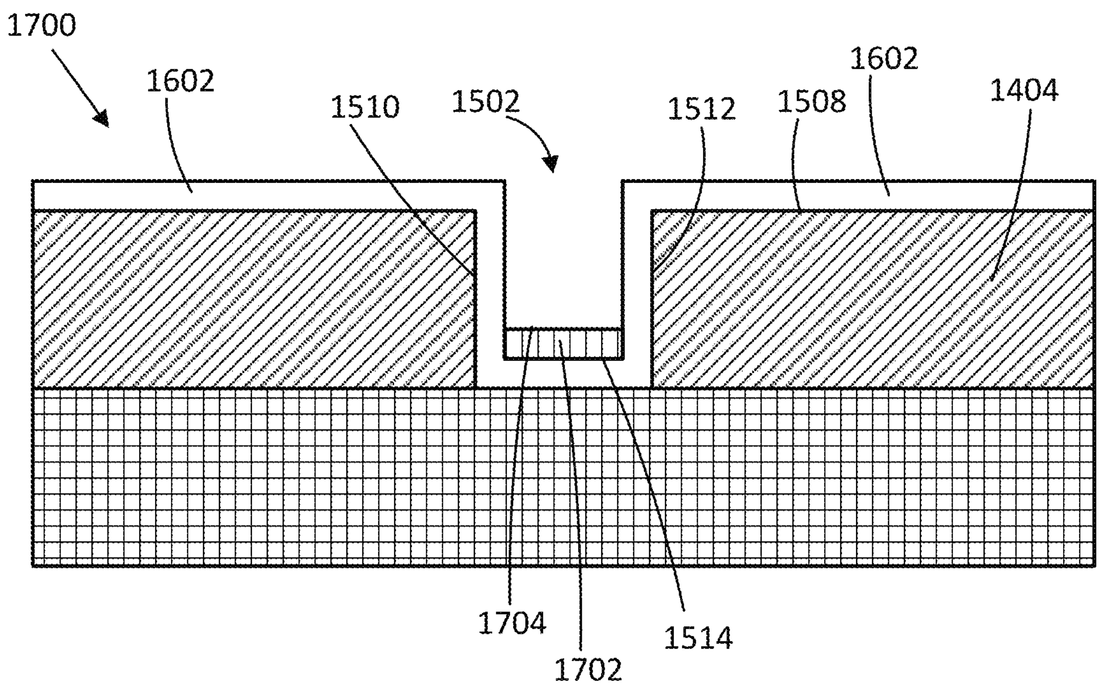


FIGURE 17

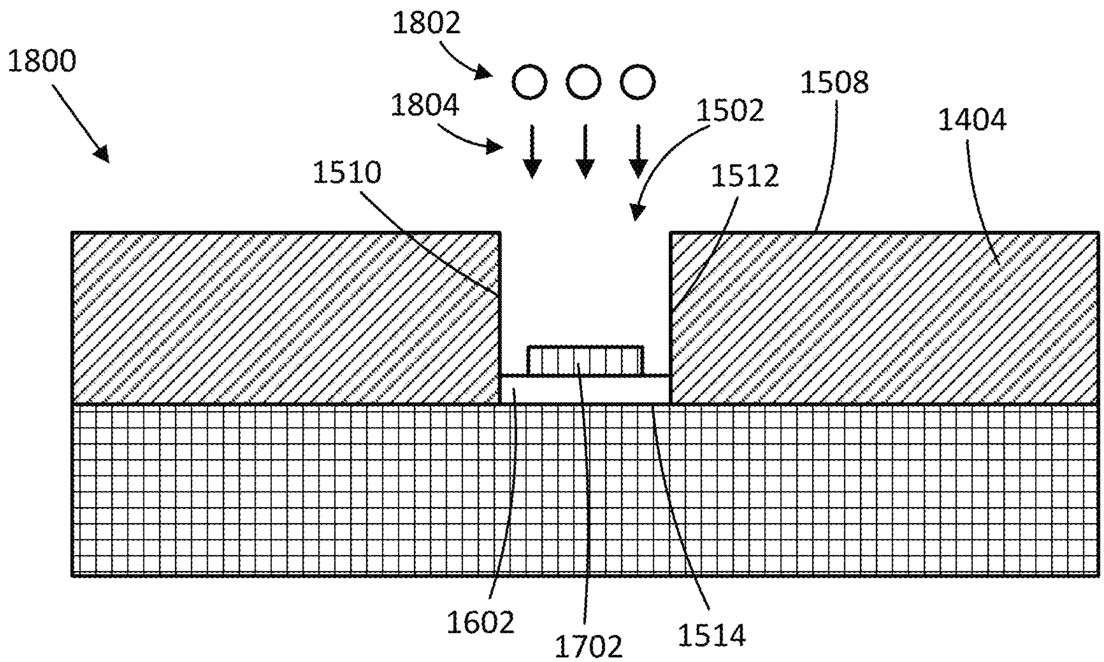


FIGURE 18

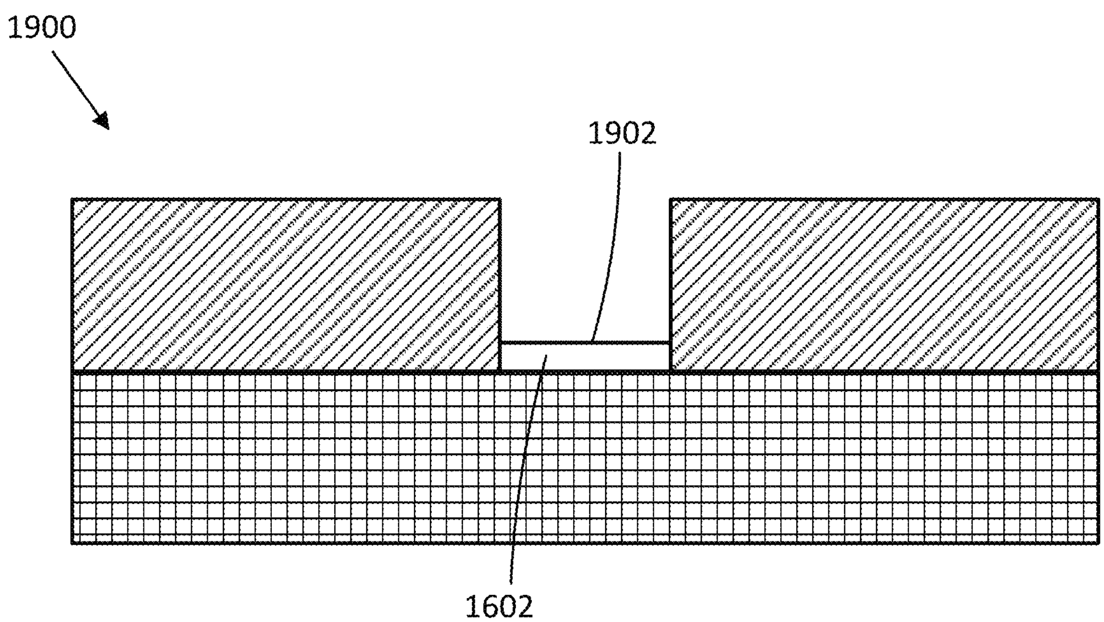


FIGURE 19

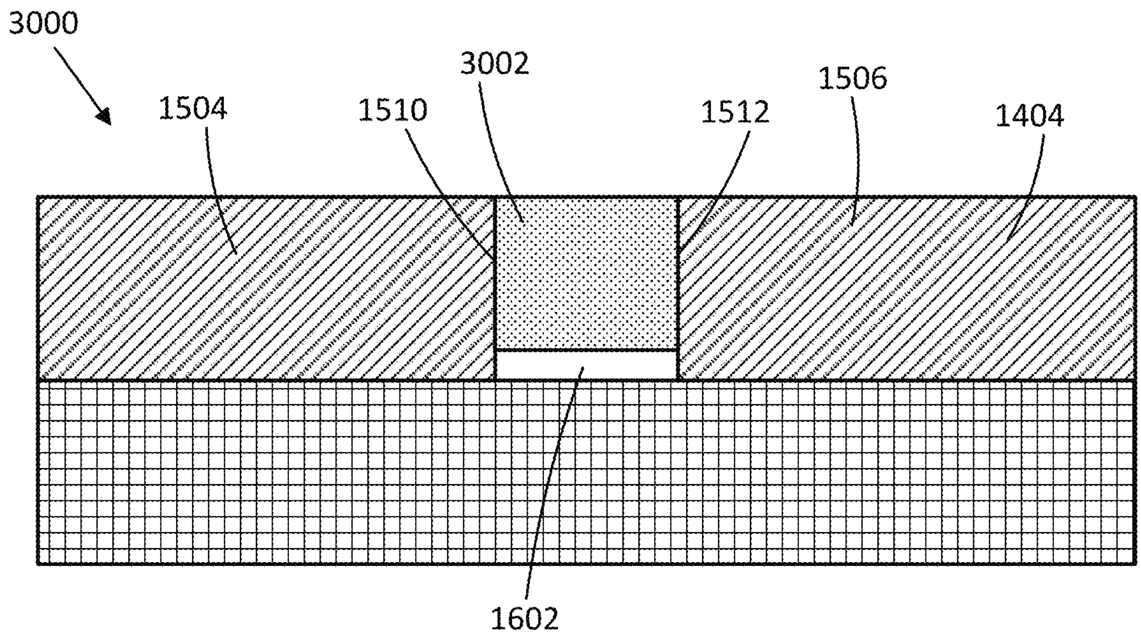


FIGURE 20

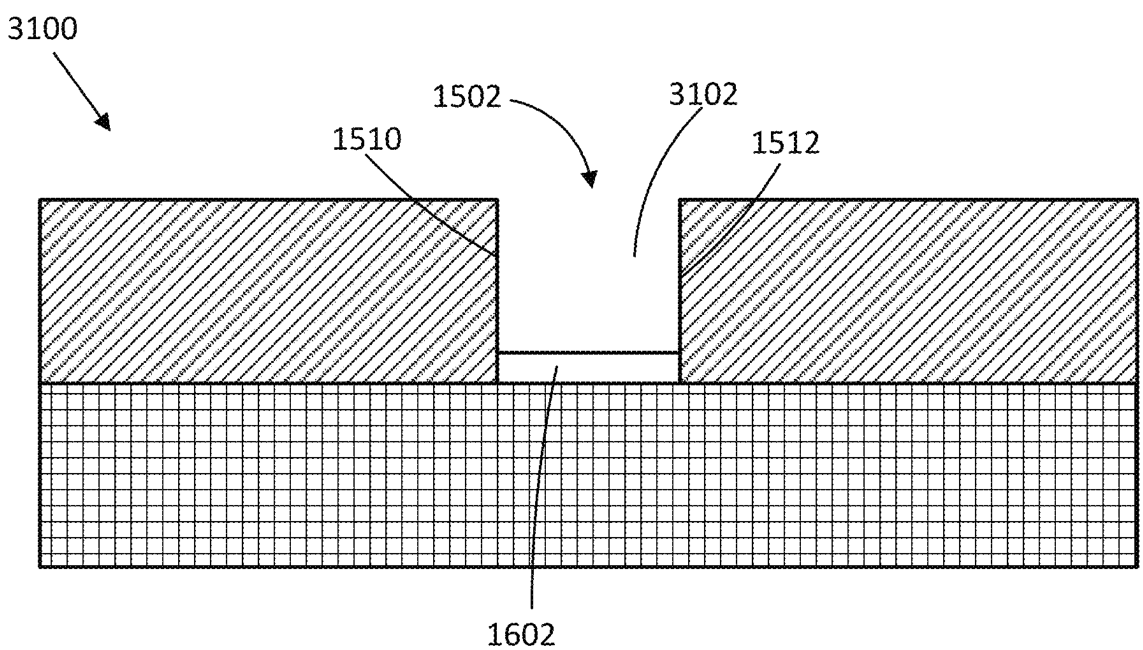


FIGURE 21

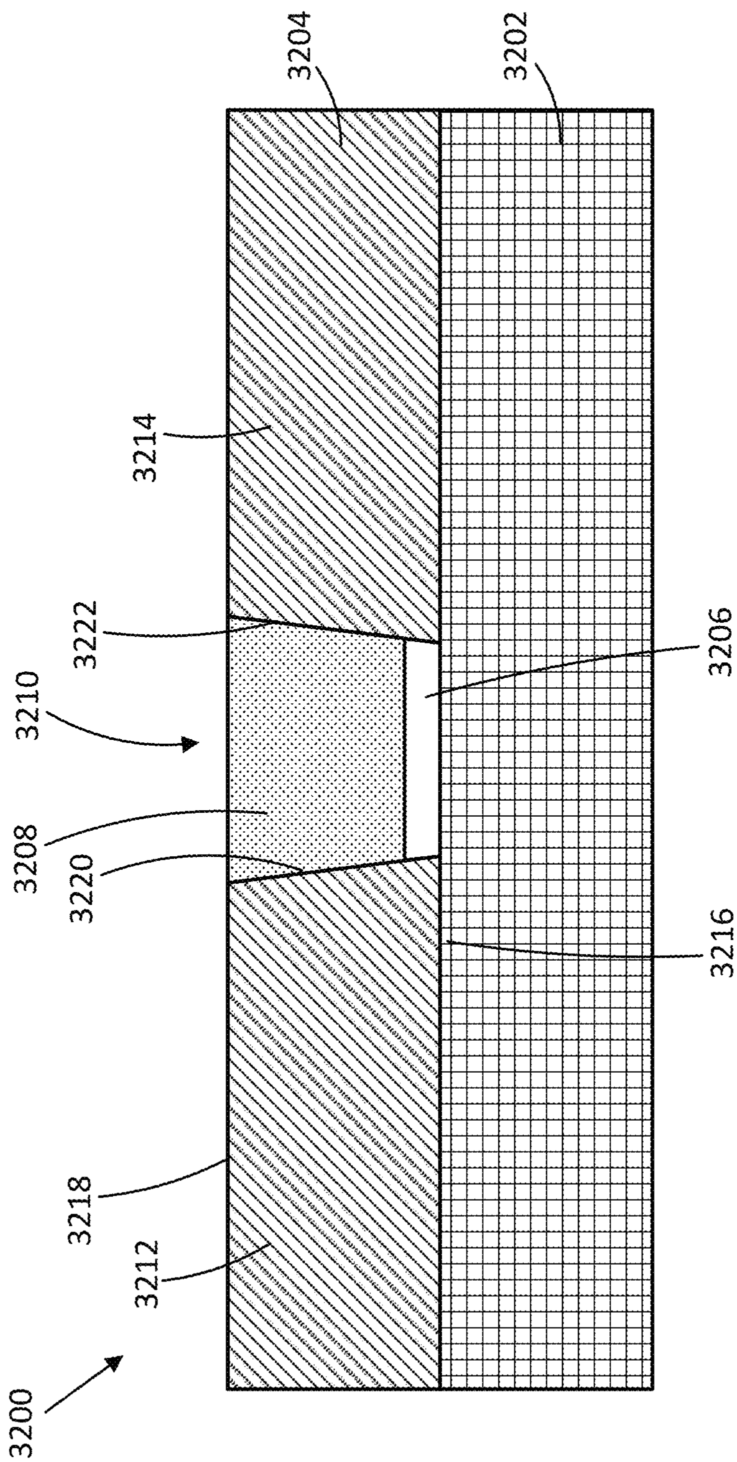


FIGURE 22

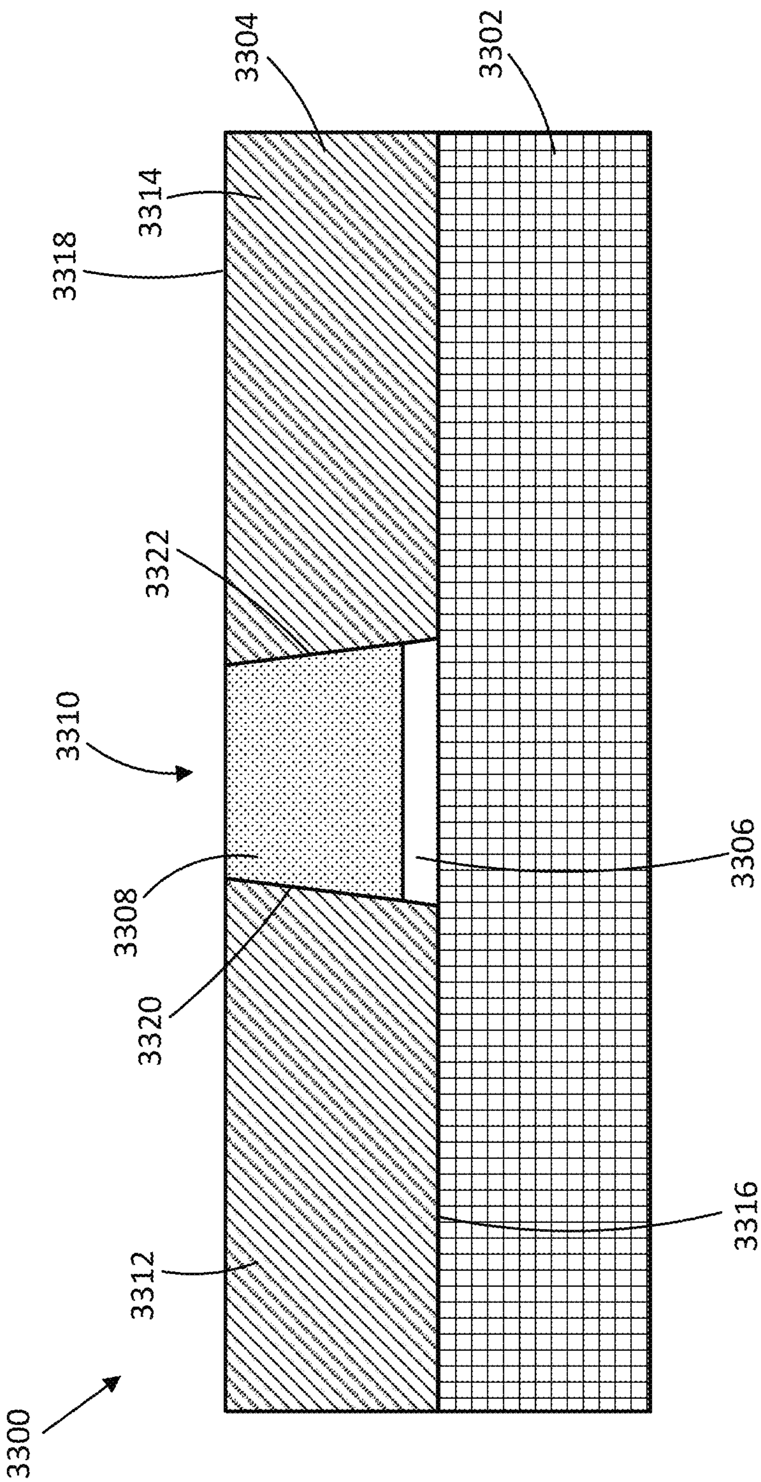


FIGURE 23

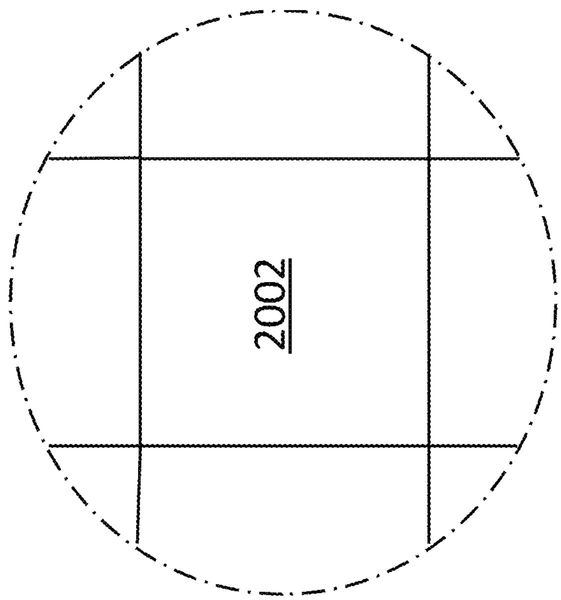


FIGURE 24B

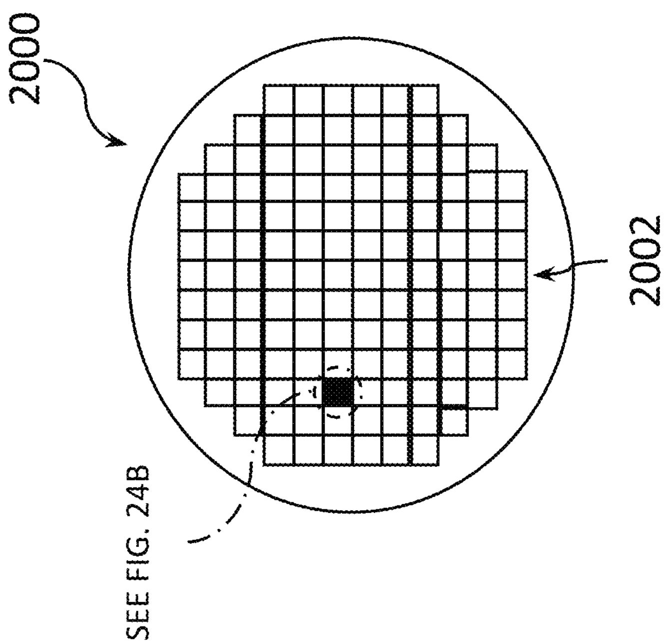
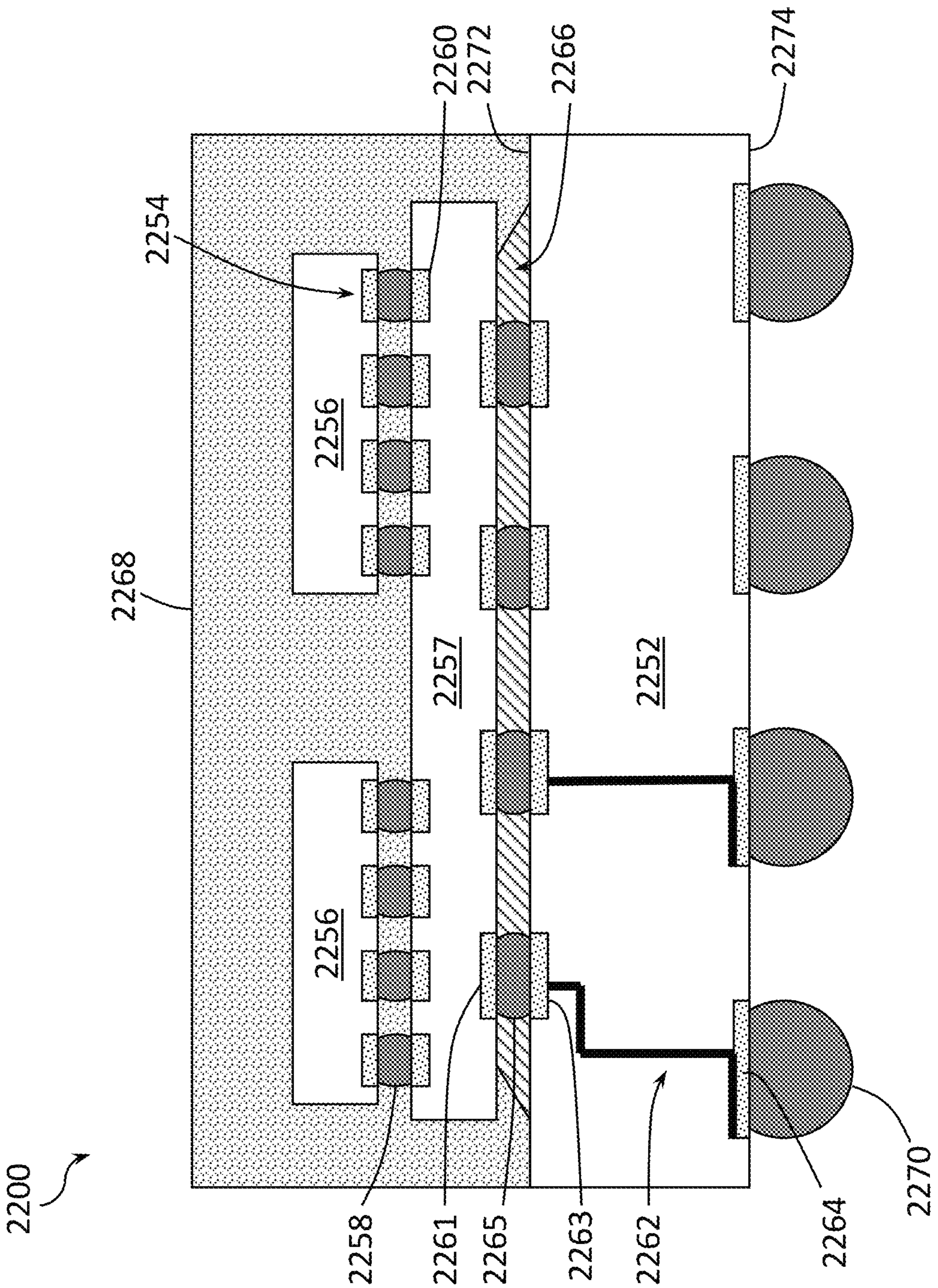


FIGURE 24A


2200
2254
2256
2256
2257
2252
2258
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2261
2262
2263
2264
2265
2266
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2272
2274

FIGURE 25

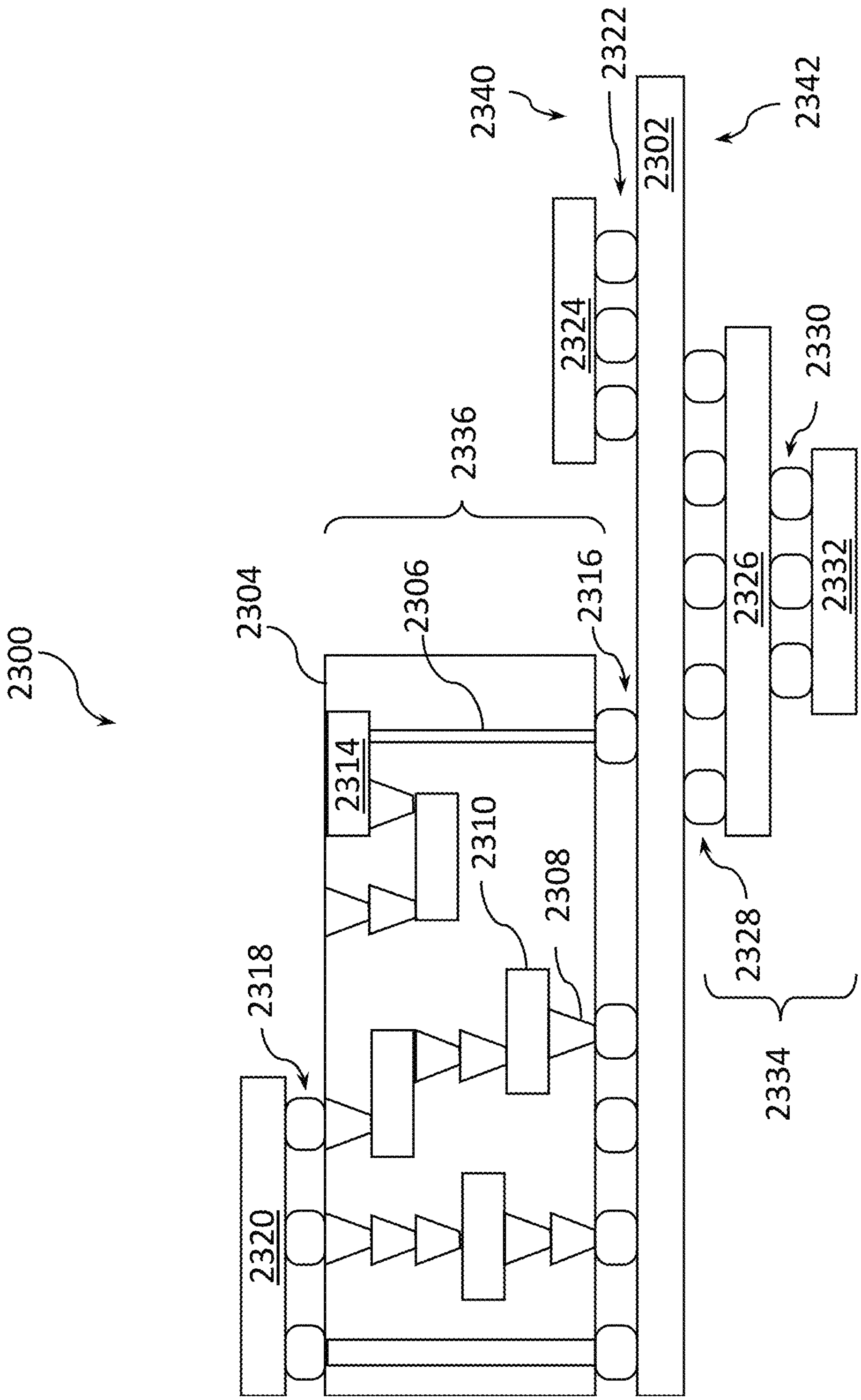
2300
2304
2306
2336
2316
2340
2322
2302
2342
2324
2330
2326
2332
2314
2310
2308
2318
2320
2328
2334

FIGURE 26

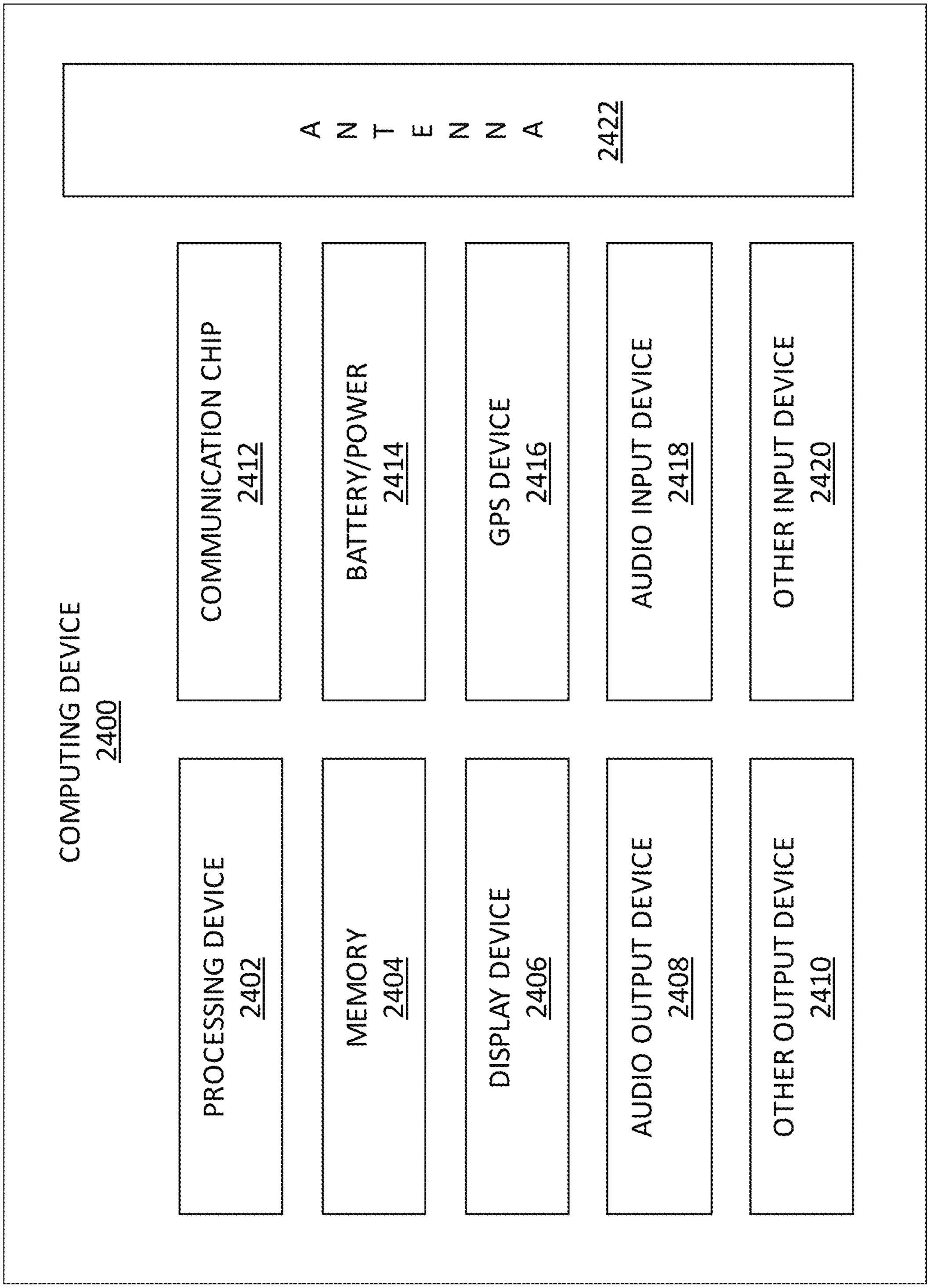
COMPUTING DEVICE
2400
PROCESSING DEVICE
2402
COMMUNICATION CHIP
2412
MEMORY
2404
BATTERY/POWER
2414
DISPLAY DEVICE
2406
GPS DEVICE
2416
AUDIO OUTPUT DEVICE
2408
AUDIO INPUT DEVICE
2418
OTHER OUTPUT DEVICE
2410
OTHER INPUT DEVICE
2420
ANTENNA
2422

FIGURE 27

INTEGRATED CIRCUIT STRUCTURE WITH FILLED RECESSES

BACKGROUND

Production of integrated circuit structures often involves producing one or more recesses in a material of the integrated circuit structure and filling the recesses with another material.

BRIEF DESCRIPTION OF THE DRAWINGS

Embodiments will be readily understood by the following detailed description in conjunction with the accompanying drawings. To facilitate this description, like reference numerals designate like structural elements. Embodiments are illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawings.

FIG. **1** illustrates an integrated circuit (IC) structure, according to some embodiments of the present disclosure.

FIG. **2** illustrates a procedure for producing IC structures, according to some embodiments of the present disclosure.

FIG. **3** illustrates an IC structure according to stage **202** of the procedure of FIG. **2**, according to some embodiments of the present disclosure.

FIG. **4** illustrates an IC structure according to stage **204** of the procedure of FIG. **2**, according to some embodiments of the present disclosure.

FIG. **5** illustrates an IC structure according to stage **206** of the procedure of FIG. **2**, according to some embodiments of the present disclosure.

FIG. **6** illustrates an IC structure according to stage **208** of the procedure of FIG. **2**, according to some embodiments of the present disclosure.

FIG. **7** illustrates another IC structure according to stage **208** of the procedure of FIG. **2**, according to some embodiments of the present disclosure.

FIG. **8** illustrates another IC structure according to stage **208** of the procedure of FIG. **2**, according to some embodiments of the present disclosure.

FIG. **9** illustrates an IC structure according to stage **212** of the procedure of FIG. **2**, according to some embodiments of the present disclosure.

FIG. **10** illustrates an IC structure according to stage **214** of the procedure of FIG. **2**, according to some embodiments of the present disclosure.

FIG. **11** illustrates an IC structure according to stage **210** of the procedure of FIG. **2**, according to some embodiments of the present disclosure.

FIG. **12** illustrates another IC structure according to stage **212** of the procedure of FIG. **2**, according to some embodiments of the present disclosure.

FIG. **13** illustrates another procedure for producing IC structures, according to some embodiments of the present disclosure.

FIG. **14** illustrates an IC structure according to stage **1302** of the procedure of FIG. **13**, according to some embodiments of the present disclosure.

FIG. **15** illustrates an IC structure according to stage **1304** of the procedure of FIG. **13**, according to some embodiments of the present disclosure.

FIG. **16** illustrates an IC structure according to stage **1306** of the procedure of FIG. **13**, according to some embodiments of the present disclosure.

FIG. **17** illustrates an IC structure according to stage **1308** of the procedure of FIG. **13**, according to some embodiments of the present disclosure.

FIG. **18** illustrates an IC structure according to stage **1310** of the procedure of FIG. **13**, according to some embodiments of the present disclosure.

FIG. **19** illustrates an IC structure according to stage **1312** of the procedure of FIG. **13**, according to some embodiments of the present disclosure.

FIG. **20** illustrates an IC structure according to stage **1316** of the procedure of FIG. **13**, according to some embodiments of the present disclosure.

FIG. **21** illustrates an IC structure according to stage **1318** of the procedure of FIG. **13**, according to some embodiments of the present disclosure.

FIG. **22** illustrates another IC structure, according to some embodiments of the present disclosure.

FIG. **23** illustrates another IC structure, according to some embodiments of the present disclosure.

FIG. **24**A is top view of a wafer that may include one or more IC structures having recesses processed via selective growth in accordance with any of the embodiments disclosed herein.

FIG. **24**B is a top view of dies that may include one or more IC structures having recesses processed via selective growth in accordance with any of the embodiments disclosed herein.

FIG. **25** is a side, cross-sectional view of an example IC package that may include one or more IC structures having one or more recesses processed via selective growth in accordance with any of the embodiments disclosed herein.

FIG. **26** is a cross-sectional side view of an IC device assembly that may include components having one or more IC structures implementing one or more recesses processed via selective growth in accordance with any of the embodiments disclosed herein.

FIG. **27** is a block diagram of an example computing device that may include one or more components with one or more IC structures having one or more recesses processed via selective growth in accordance with any of the embodiments disclosed herein.

DETAILED DESCRIPTION

Overview

Disclosed herein are integrated circuit (IC) structures, packages, and device assemblies that include a recess where materials within the recess, or lack thereof, are produced with a bottom-up growth procedure, which may be referred to as selective growth. Further disclosed herein are procedures incorporating angled etch or sacrificial material to produce the IC structures, packages, and device assemblies with the recess. In particular, an IC structure includes a recess formed in a dielectric material of the IC structure. A first material is disposed on the bottom and the sidewalls of the recess. An angled etch procedure, or a sacrificial material and etch procedure can be performed to remove at least a portion of the first material from the sidewalls of the recess, thereby exposing at least a portion of each of the sidewalls of the recess. A bottom-up growth procedure is performed to produce a second material on the first material, where the bottom-up growth procedure avoids producing the seams or voids that can be produced when none of the first material is removed from the sidewalls of the recess.

In one aspect of the present disclosure, an example integrated circuit (IC) structure includes a first dielectric material, a second dielectric material on the first dielectric material, and a recess in the second dielectric material, wherein the recess includes a bottom, a top, and sidewalls. The IC further includes a first material within the recess and at a bottom of the recess, wherein the first material includes a metal and oxygen, a self-assembled monolayer (SAM) material, or an organic material, and a second material within the recess and between the first material and the top of the recess, wherein the second material is in contact with the sidewalls of the recess.

Each of the structures, packages, methods, devices, and systems of the present disclosure may have several innovative aspects, no single one of which being solely responsible for the all of the desirable attributes disclosed herein. Details of one or more implementations of the subject matter described in this specification are set forth in the description below and the accompanying drawings.

In the following detailed description, various aspects of the illustrative implementations may be described using terms commonly employed by those skilled in the art to convey the substance of their work to others skilled in the art. For example, the term "connected" means a direct electrical or magnetic connection between the things that are connected, without any intermediary devices, while the term "coupled" means either a direct electrical or magnetic connection between the things that are connected, or an indirect connection through one or more passive or active intermediary devices. The term "circuit" means one or more passive and/or active components that are arranged to cooperate with one another to provide a desired function. If used, the terms "oxide," "carbide," "nitride," etc. refer to compounds containing, respectively, oxygen, carbon, nitrogen, etc. Similarly, the terms naming various compounds refer to materials having any combination of the individual elements within a compound (e.g., "gallium nitride" or "GaN" refers to a material that includes gallium and nitrogen, "aluminum indium gallium nitride" or "AlInGaN" refers to a material that includes aluminum, indium, gallium and nitrogen, and so on). Further, the term "high-k dielectric" refers to a material having a higher dielectric constant (k) than silicon oxide, while the term "low-k dielectric" refers to a material having a lower k than silicon oxide. The terms "substantially," "close," "approximately," "near," and "about," generally refer to being within +/−20%, preferably within +/−10%, of a target value based on the context of a particular value as described herein or as known in the art. Similarly, terms indicating orientation of various elements, e.g., "coplanar," "perpendicular," "orthogonal," "parallel," or any other angle between the elements, generally refer to being within +/−5-20% of a target value based on the context of a particular value as described herein or as known in the art.

The terms "over," "under," "between," "on," and "abut" as used herein refer to a relative position of one material layer or component with respect to other layers or components. For example, one layer located over or under another layer may be directly in contact with the other layer or may have one or more intervening layers. Moreover, one layer located between two layers may be directly in contact with one or both of the two layers or may have one or more intervening layers. A first layer described to be "on" a second layer refers to the first layer that is in direct contact with that second layer or has one or more intervening layers. In contrast, a first layer described to be "abutting" a second layer refers to the first layer that is in direct contact with the that second layer. Similarly, unless explicitly stated otherwise, one feature disposed between two features may be in direct contact with the adjacent features or may have one or more intervening layers.

For the purposes of the present disclosure, the phrase "A and/or B" means (A), (B), or (A and B). For the purposes of the present disclosure, the phrase "A, B, and/or C" means (A), (B), (C), (A and B), (A and C), (B and C), or (A, B, and C). The term "between," when used with reference to measurement ranges, is inclusive of the ends of the measurement ranges. As used herein, the notation "A/B/C" means (A), (B), and/or (C).

The description uses the phrases "in an embodiment" or "in embodiments," which may each refer to one or more of the same or different embodiments. Furthermore, the terms "comprising," "including," "having," and the like, as used with respect to embodiments of the present disclosure, are synonymous. The disclosure may use perspective-based descriptions such as "above," "below," "top," "bottom," and "side"; such descriptions are used to facilitate the discussion and are not intended to restrict the application of disclosed embodiments. The accompanying drawings are not necessarily drawn to scale. Unless otherwise specified, the use of the ordinal adjectives "first," "second," and "third," etc., to describe a common object, merely indicate that different instances of like objects are being referred to, and are not intended to imply that the objects so described must be in a given sequence, either temporally, spatially, in ranking or in any other manner.

In the following detailed description, reference is made to the accompanying drawings that form a part hereof, and in which is shown, by way of illustration, embodiments that may be practiced. It is to be understood that other embodiments may be utilized, and structural or logical changes may be made without departing from the scope of the present disclosure. Therefore, the following detailed description is not to be taken in a limiting sense. For convenience, if a collection of drawings designated with different letters are present, e.g., FIGS. **5**A-**5**B, such a collection may be referred to herein without the letters, e.g., as "FIG. **5**." In the drawings, same reference numerals refer to the same or analogous elements/materials shown so that, unless stated otherwise, explanations of an element/material with a given reference numeral provided in context of one of the drawings are applicable to other drawings where element/materials with the same reference numerals may be illustrated.

In the drawings, some schematic illustrations of example structures of various structures, devices, and assemblies described herein may be shown with precise right angles and straight lines, but it is to be understood that such schematic illustrations may not reflect real-life process limitations which may cause the features to not look so "ideal" when any of the structures described herein are examined using e.g., scanning electron microscopy (SEM) images or transmission electron microscope (TEM) images. In such images of real structures, possible processing defects could also be visible, e.g., not-perfectly straight edges of materials, tapered vias or other openings, inadvertent rounding of corners or variations in thicknesses of different material layers, occasional screw, edge, or combination dislocations within the crystalline region(s), and/or occasional dislocation defects of single atoms or clusters of atoms. There may be other defects not listed here but that are common within the field of device fabrication.

Various operations may be described as multiple discrete actions or operations in turn in a manner that is most helpful in understanding the claimed subject matter. However, the order of description should not be construed as to imply that these operations are necessarily order dependent. In particular, these operations may not be performed in the order of presentation. Operations described may be performed in a different order from the described embodiment. Various additional operations may be performed, and/or described operations may be omitted in additional embodiments.

In various embodiments, components associated with an IC include, for example, transistors, diodes, power sources, resistors, capacitors, inductors, sensors, transceivers, receivers, antennas, etc. Components associated with an IC may include those that are mounted on an IC, provided as an integral part of an IC, or those connected to an IC. The IC may be either analog or digital and may be used in a number of applications, such as microprocessors, optoelectronics, logic blocks, audio amplifiers, etc., depending on the components associated with the IC. In some embodiments, IC structures as described herein may be included in a RFIC, which may, e.g., be included in any component associated with an IC of an RF receiver, an RF transmitter, or an RF transceiver, e.g., as used in telecommunications within base stations (BS) or user equipment (UE). Such components may include, but are not limited to, power amplifiers, low-noise amplifiers, RF filters (including arrays of RF filters, or RF filter banks), switches, upconverters, downconverters, and duplexers. In some embodiments, the IC structures as described herein may be employed as part of a chipset for executing one or more related functions in a computer.

A legacy approach to forming material within a recess of an IC structures includes depositing a conformal metal oxide layer on an entirety of the bottom and the sidewalls of the recess. Atomic layer deposition can then be utilized to form another material on the conformal metal oxide layer to fill the rest of the recess. However, the deposition of the material with the conformal metal oxide layer on the entirety of the sidewalls often results in seams or voids being formed in the material, where the material is absent in the seams or voids. The seams or voids can be sources of defects in the IC structures. For example, subsequent metal processing may penetrate into the seams and cause defects in the IC structures.

FIG. **1** illustrates an IC structure **100**, according to some embodiments of the present disclosure. The IC structure **100** can be implemented in an IC die in some embodiments. The IC structure **100** may undergo further processing in IC die production where additional materials can be affixed on the top and/or bottom of the IC structure **100**. A further IC component may be coupled to the IC die with the IC structure **100** in some embodiments.

The IC structure **100** includes a first dielectric material **102**. The first dielectric material **102** can form a layer in some embodiments of the IC structure **100**. The first dielectric material **102** may comprise a material with a high dielectric constant, such as a dielectric constant value greater than 3.9. The first dielectric material **102** may comprise a dielectric material including oxygen. For example, the first dielectric material **102** may comprise silicon dioxide (which includes silicon and oxygen), hafnium dioxide (which includes hafnium and oxygen), zirconium dioxide (which includes zirconium and oxygen), titanium dioxide (which includes titanium and oxygen), aluminum oxide (which includes aluminum and oxygen), or some combination thereof.

The IC structure **100** further includes a second dielectric material **104**. The second dielectric material **104** is located on the first dielectric material **102**. The first dielectric material **102** and the second dielectric material **104** may be referred to as a base assembly. The second dielectric material **104** may abut a top surface **108** of the first dielectric material **102**. The second dielectric material **104** may comprise a material with a high dielectric constant, such as a dielectric constant value greater than 3.9. The second dielectric material **104** may comprise a dielectric material including oxygen or a dielectric material including nitrogen. For example, the second dielectric material **104** may comprise silicon dioxide (which includes silicon and oxygen), hafnium dioxide (which includes hafnium and oxygen), zirconium dioxide (which includes zirconium and oxygen), titanium dioxide (which includes titanium and oxygen), aluminum oxide (which includes aluminum and oxygen), aluminum nitride (which includes aluminum and nitrogen), silicon nitride (which includes silicon and nitrogen), or some combination thereof. In some embodiments, the first dielectric material **102** and the second dielectric material **104** may comprise a same dielectric material, which can be a dielectric material including oxygen or a dielectric material including nitrogen.

The second dielectric material **104** comprises a first portion **104***a* and a second portion **104***b*. A recess **106** is located between the first portion **104***a* of the second dielectric material **104** and the second portion **104***b* of the second dielectric material **104**, where the second dielectric material **104** is absent from the recess **106**. The recess **106** can extend from a top surface **114** of the second dielectric material **104** to the top surface **108** of the first dielectric material **102**. In other embodiments, the recess can extend from the top surface **114** of the second dielectric material **104** through a portion of the second dielectric material **104** or through the second dielectric material **104** and a portion of the first dielectric material **102**. The first portion **104***a* of the second dielectric material **104** abuts the recess **106** on a first side and the second portion **104***b* of the second dielectric material **104** abuts the recess **106** on a second side, the second side of the recess **106** being opposite to the first side. In these embodiments, the first portion **104***a* of the second dielectric material **104** defines a first sidewall **110** of the recess **106** and the second portion **104***b* of the second dielectric material **104** defines a second sidewall **112** of the recess **106**. In other embodiments, another material may be located between the first portion **104***a* of the second dielectric material **104** and the recess **106**, and between the second portion **104***b* of the second dielectric material **104** and the recess **106**, as described further throughout the disclosure. In these other embodiments, the other material may define the first sidewall **110** of the recess **106** and the second sidewall **112** of the recess **106**.

The IC structure **100** further includes a first material **116** located within the recess **106**. The first material may comprise a metal oxide material (such as silicon oxide), a self-assembled monolayer (SAM) material, or an organic material in some embodiments. In some embodiments, the first material **116** may comprise mono-dialkylaminosilanes, bis-dialkylaminosilanes, and tris-dialkylaminosilanes (such Me2N—SiR3, (Me2N)2SiR2, (Me2N)3SiR, where R is a small alkyl such as Me, Et, Pr, or nBu), small alkyl chains or fluorinated chains and head groups (including alkoxysilanes, aminosilanes (such as (Me2N)SiMe2-SiMe2 (NMe2))), chlorosilanes; alkenes, alkynes, amines, phosphines, thiols, phosphonic acids, carboxylic acids, chlorosilanes, fluorosilanes, bromosilanes, alkoxysilanes, phosphonic acids with small chains, phosphonates with small chains, or some combination. In some embodiments, the first material **116** may comprise small alkyl chains or fluorinated chains, including alkoxysilanes, and chlorosilanes. Further, the first material **116** may be phosphonic acids and/or phosphonates in some embodiments. The first material **116** is located at a bottom **118** of the recess **106** and between the first sidewall **110** and the second sidewall **112**.

The first material **116** can abut a portion of the first sidewall **110** and a portion of the second sidewall **112**, where the portion of the first sidewall **110** is less than an entirety of the first sidewall **110** and the portion of the second sidewall **112** is less than an entirety of the second sidewall **112**. In other embodiments, first material **116** can abut only one of the first sidewall **110** and the second sidewall **112**, or may not abut either of the first sidewall **110** and the second sidewall **112**. In the illustrated embodiment, the first material **116** is illustrated with a uniform thickness across the recess **106**. In other embodiments, the thickness of the first material **116** may vary across the recess **106**.

The IC structure **100** further include a second material **120** located within the recess **106**. The second material **120** may comprise a dielectric material, (such as silicon oxide, silicon nitride, or metal oxides, including ZrO2, HfO2, and Al2O3) where the dielectric material may act as a plug. In other embodiments, the second material **120** may comprise a metal (such as Ti, TiN, Ta, TaN, Al, Cu, Co, Ru, or W), where the metal may act as a via-like structure. The second material **120** is located on the first material **116**, where the first material **116** is located between the second material **120** and the first dielectric material **102**. Further, the second material **120** is located between the first sidewall **110** and the second sidewall **112**. The second material **120** abuts a portion of the first sidewall **110** and a portion of the second sidewall **112**. In embodiments where the first material **116** abuts a portion of the first sidewall **110** and/or a portion of the second sidewall **112**, the first material **116** abuts a first portion of the first sidewall **110** and a first portion of the second sidewall **112** and the second material **120** abuts a second portion of the first sidewall **110** and a second portion of the second sidewall **112**, where the first material **116** and the second material **120** may collectively abut an entirety of the first sidewall **110** and an entirety of the second sidewall **112**. In embodiments where the first material **116** does not abut either of the first sidewall **110** and the second sidewall **112**, the second material **120** may abut an entirety of the first sidewall **110** and an entirety of the second sidewall **112**. The first material **116** and the second material **120** collectively fill the recess **106**, where a top surface **122** of the second material **120** is substantially aligned with the top surface **114** of the second dielectric material **104**.

FIG. **2** illustrates a procedure **200** for producing IC structures, according to some embodiments of the present disclosure. FIG. **3** through FIG. **11** illustrates IC structures corresponding to different stages of the procedure **200**.

The procedure **200** can begin with a formed first dielectric material, such as the first dielectric material **302** illustrated in FIG. **3**. The first dielectric material may include one or more of the features of the first dielectric material **102** (FIG. **1**), including the materials of which the first dielectric material **102** may be comprised. The procedure **200** initiates with stage **202**. In stage **202**, a second dielectric material is applied to the first dielectric material. The second dielectric material may include one or more of the features of the second dielectric material **104** (FIG. **1**), including the materials of which the second dielectric material **104** may be comprised. In particular, the second dielectric material is applied to a top surface of the first dielectric material. The second dielectric material can be applied at approximately a uniform thickness across the top surface of the first dielectric material and can extend for an entirety of the top surface of the first dielectric material. The procedure **200** may proceed from stage **202** to stage **204**.

FIG. **3** illustrates an IC structure **300** according to stage **202** of the procedure **200** of FIG. **2**, according to some embodiments of the present disclosure. The IC structure **300** includes a first dielectric material **302**. The IC structure **300** further includes a second dielectric material **304** located on the first dielectric material **302**. In particular, the second dielectric material **304** is located on a top surface **306** of the first dielectric material **302**. The second dielectric material **304** extends on the top surface **306** of the first dielectric material **302** for an entirety of the top surface **306**, where the second dielectric material **304** has an approximately uniform thickness for the entirety of the top surface **306**.

In stage **204**, patterning of the second dielectric material is performed. Patterning of the second dielectric material may include removing a portion of the second dielectric material from the top surface of the first dielectric material to produce a certain pattern. In some embodiments, patterning of the second dielectric material may include curing of a portion of the second dielectric material and removing the uncured portion of the second dielectric material or cured portion of the second dielectric material, depending on the chemical makeup of the second dielectric material and the removal procedure being applied. For the disclosed embodiments, the pattern produced by the patterning of the second dielectric material includes at least one recess. The patterned second dielectric material and the first dielectric material may be collectively referred to as a base assembly. The procedure **200** may proceed from stage **204** to stage **206**.

FIG. **4** illustrates an IC structure **400** according to stage **204** of the procedure **200** of FIG. **2**, according to some embodiments of the present disclosure. In particular, FIG. **4** illustrates the IC structure **400** produced by stage **204** being applied to the IC structure **300** (FIG. **3**) in accordance with some embodiments disclosed herein.

The performance of stage **204** caused a portion of the second dielectric material **304** to be removed. The removal of the portion of the second dielectric material **304** produces a recess **402** within the second dielectric material **304**, where the recess **402** is located between a first portion **404** of the second dielectric material **304** and a second portion **406** of the second dielectric material **304**. In the illustrated embodiment, the first portion **404** of the second dielectric material **304** defines a first sidewall **410** of the recess **402** and the second portion **406** of the second dielectric material **304** defines a second sidewall **412** of the recess **402**. While the first sidewall **410** and the second sidewall **412** are illustrated as being straight and perpendicular to the top surface **306** of the first dielectric material **302** in the illustrated embodiment, it is to be understood that the first sidewall **410** and/or the second sidewall **412** can be curved, and/or tapered or reentrant in other embodiments. The recess **402** extends from a top surface **408** of the second dielectric material **304** to the top surface **306** of the first dielectric material **302**, where the top surface **408** of the second dielectric material **304** includes a top surface of the first portion **404** and the second portion **406** of the second dielectric material **304**. The top surface **306** of the first dielectric material **302** defines a bottom **414** of the recess **402**. In other embodiments, the recess **402** may extend partially through the second dielectric material **304**, or through the second dielectric material **304** and partially through the first dielectric material **302**.

In stage **206**, a first material is formed on the IC structure. The first material may include one or more of the features of the first material **116** (FIG. **1**), including the materials of which the first material **116** may be comprised. In particular, the first material is formed on the top surface of the second dielectric material and within the recess at the sidewalls and the bottom of the recess. The first material may be deposited on the top surface of the second dielectric material and within the recess at the sidewalls and the bottom of the recess via a deposition procedure. The first material has a substantially uniform thickness from the surface to which it is applied (i.e., the top surface of the second dielectric material, the sidewalls of the recess, or the bottom of the recess). The procedure **200** may proceed from stage **206** to stage **208**.

FIG. **5** illustrates an IC structure **500** according to stage **206** of the procedure **200** of FIG. **2**, according to some embodiments of the present disclosure. In particular, FIG. **5** illustrates the IC structure **500** produced by stage **206** being applied to the IC structure **400** (FIG. **4**) in accordance with some embodiments disclosed herein.

The IC structure **500** includes a first material **502** that has been formed on the top surface **408** of the second dielectric material **304**, at the first sidewall **410** and the second sidewall **412** within the recess **402**, and at the bottom **414** of the recess **402**. The first material **502** abuts the top surface **408** of the second dielectric material **304**, the first sidewall **410**, the second sidewall **412**, and the bottom **414** of the recess **402**. The first material **502** has a substantially uniform thickness from the surfaces on which the first material **502** has been formed.

In stage **208**, an angled etch may be performed on the IC structure. The angled etch may remove a portion of the first material formed on the IC structure in stage **206**. The angled etch includes directing ions at the IC structure, where the ions are directed at a non-perpendicular angle to the bottom of the recess. In some embodiments, the angle of the ions relative to the bottom of the recess can be less than 90 degrees and greater than or equal to 60 degrees. In some embodiments, the ions may be directed at multiple angles within a predetermined distribution of angles, where the distribution can be less than 90 degrees and greater than or equal to 60 degrees in some embodiments. The angled etch can cause the portion of the first material contacted by the ions to be removed from the IC structure. In some embodiments, the ions may comprise fluorine (F) ions, where the angled etch utilizes an F+-based chemistry. In other embodiments, the ions may comprise ions of other materials utilized for etching procedures, such as chlorine (Cl) ions in a Cl+-based chemistry or bromine (Br) ions in a Br+-based chemistry. In some embodiments, traces of the ions may remain on the IC structure after the angled etch has been performed. A de-scum or de-ash procedure may be performed on the IC structure to remove the traces of the ions in some embodiments.

FIG. **6** illustrates an IC structure **600** according to stage **208** of the procedure **200** of FIG. **2**, according to some embodiments of the present disclosure. In particular, FIG. **6** illustrates the IC structure **600** produced by stage **208** being applied to the IC structure **500** (FIG. **5**) in accordance with some embodiments disclosed herein.

FIG. **6** illustrates ions **602** that are directed at the IC structure **600**. Arrows **604** illustrate directions of travel of the ions **602**. The ions **602** are directed at the top surface **408** of the second dielectric material **304** and are directed at angle to the bottom **414** of the recess **402**. The ions **602** can contact the first material **502** located on the top surface **408** of the second dielectric material **304** and the first material **502** located on a portion of the first sidewall **410** and the second sidewall **412** (collectively referred to as "the sidewalls"). Due to the angle at which the ions **602** are directed and the second dielectric material **304**, the ions **602** contact the first material **502** on the portion of the sidewalls and removes the first material **502** from the portion of the sidewalls, but blocks the ions **602** from contacting the first material **502** at the bottom **414** of the recess **402** leaving the first material **502** at the bottom **414** of the recess **402**. It should be understood that the ions **602** and the arrows **604** illustrated are to illustrate the features of the angled etch, and that more ions **602** than illustrated can be directed at the IC structure **600**.

The IC structure **600** illustrated is a result of stage **208** in some embodiments. In particular, the IC structure **600** illustrated can result from the ions **602** being directed such that the ions **602** contact the portion of the sidewalls above a top surface **606** of the remaining first material **502** as illustrated. The first material **502** has been removed by the ions **602** from the top surface **408** of the second dielectric material **304** and portion of the sidewalls above the top surface **606** of the remaining first material **502**. Accordingly, the first material **502** is located within the recess **402** and at the bottom **414** of the recess **402** in the illustrated embodiment. Further, the first material **502** is located on the top surface **306** of the first material **502**. The first material **502** extends from the first sidewall **410** to the second sidewall **412** and abuts both the first sidewall **410** and the second sidewall **412**. Further, the first material **502** has approximately a uniform thickness in the illustrated embodiment.

FIG. **7** illustrates another IC structure **700** according to stage **208** of the procedure **200** of FIG. **2**, according to some embodiments of the present disclosure. In particular, FIG. **7** illustrates the IC structure **700** produced by stage **208** being applied to the IC structure **500** (FIG. **5**) in accordance with some embodiments disclosed herein.

FIG. **7** illustrates ions **702** being directed at the IC structure **700**. Arrows **704** illustrate the direction at which the ions **702** are directed at the IC structure **700**. The ions **702** are directed at a top surface **408** of the second dielectric material **304** and are directed at an angle to the bottom **414** of the recess **402**. In contrast to the embodiment illustrated in FIG. **6**, the angle at which the ions **702** are directed to the bottom **414** of the recess **402** is greater than the angle at which the ions **602** (FIG. **6**) are directed to the bottom **414** of the recess **402**.

The ions **702** can contact the first material **502** located on the top surface **408** of the second dielectric material **304**, the first material **502** located on the first sidewall **410** and the second sidewall **412** (collectively referred to as "the sidewalls"), and a first portion of the first material **502** located at the bottom **414** of the recess **402**. Due to the angle at which the ions **702** are directed and the second dielectric material **304**, the ions **702** contact the first material **502** on the sidewalls and a first portion of the first material **502** at a bottom **414** of the recess, and removes the first material **502** from the sidewalls and the first portion located at the bottom **414** of the recess **402**. The ions **702** may further avoid contact with a second portion of the first material **502** located at the bottom **414** of the recess **402** and leaves the second portion of the first material **502** located at the bottom **414** of the recess **402**. It should be understood that the ions **702** and the arrows **704** illustrated are to illustrate the features of the angled etch, and that more ions **702** than illustrated can be directed at the IC structure **700**.

The IC structure **700** illustrated is a result of stage **208** in some embodiments. In particular, the IC structure **700** illustrated results from the ions **702** being directed such that the ions **702** contact the sidewalls and a portion of the bottom **414** of the recess **402**. The first material **502** has been removed by the ions **702** from the top surface **408** of the second dielectric material **304**, the sidewalls of the recess **402**, and the portion of the bottom **414** of the recess **402**.

Accordingly, the first material **502** is located within the recess **402** and at the bottom **414** of the recess **402** in the illustrated embodiment. Further, the first material **502** is located on the top surface **306** of the first material **502**. The first material **502** extends between the first sidewall **410** to the second sidewall **412** without contacting either of the first sidewall **410** and the second sidewall **412**.

FIG. **8** illustrates another IC structure **800** according to stage **208** of the procedure **200** of FIG. **2**, according to some embodiments of the present disclosure. In particular, FIG. **8** illustrates the IC structure **800** produced by stage **208** being applied to the IC structure **500** (FIG. **5**) in accordance with some embodiments disclosed herein.

FIG. **8** illustrates ions **802** being directed at the IC structure **800**. Arrows **804** illustrate the direction at which the ions **802** are directed at the IC structure **800**. The ions **802** are directed at a top surface **408** of the second dielectric material **304** and are directed at an angle to the bottom **414** of the recess **402**. In contrast to the embodiment illustrated in FIG. **6**, the angle at which the ions **802** are directed to the bottom **414** of the recess **402** is less than the angle at which the ions **602** (FIG. **6**) are directed to the bottom **414** of the recess **402**.

The ions **802** can contact the first material **502** located on the top surface **408** of the second dielectric material **304** and a portion the first material **502** located on the first sidewall **410** and the second sidewall **412** (collectively referred to as "the sidewalls"), wherein the portion of the first material **502** located on the sidewalls is greater than the portion of the first material **502** in FIG. **6**. Due to the angle at which the ions **802** are directed and the second dielectric material **304**, the ions **802** contact the portion of the first material **502** on the sidewalls, and removes the portion of the first material **502** from the sidewalls. The ions **802** may further avoid contact with a second portion of the first material **502** located on the sidewalls and at the bottom **414** of the recess **402**, and leaves the second portion of the first material **502** on the sidewalls and located at the bottom **414** of the recess **402**. It should be understood that the ions **802** and the arrows **804** illustrated are to illustrate the features of the angled etch, and that more ions **802** than illustrated can be directed at the IC structure **800**.

The IC structure **800** illustrated is a result of stage **208** in some embodiments. In particular, the IC structure **800** illustrated results from the ions **802** being directed such that the ions **802** contact a portion of the sidewalls of the recess **402**. The first material **502** has been removed by the ions **802** from the top surface **408** of the second dielectric material **304**, and the portion of the sidewalls of the recess **402**. Accordingly, the first material **502** is located within the recess **402** and at the bottom **414** of the recess **402** in the illustrated embodiment. Further, the first material **502** is located on the top surface **306** of the first material **502**. The first material **502** extends from the first sidewall **410** to the second sidewall **412** and contacts both the first sidewall **410** and the second sidewall **412**. The thickness of the first material **502** near the first sidewall **410** and the second sidewall **412** is greater than the thickness of the first material **502** further away from the first sidewall **410** and the second sidewall **412**.

While FIGS. **6** through **8** illustrate some examples of IC structures that may be produced by stage **208**, it should be understood that other resultant IC structures may be produced based on the angle of the ions relative to the bottom **414** of the recess **402** and/or the shape (ex. thickness and/or contours) of the second dielectric material **304**. Other embodiments may include any embodiments where the first material **502** has been removed from the top surface **408** of the second dielectric material **304** and at least a portion of the first material **502** on the sidewalls, while some of the first material **502** remains within the recess **402** at the bottom **414** of the recess **402**.

From stage **208**, the procedure **200** may proceed to stage **210** or stage **212** depending on the chemical makeup of the first material applied in stage **206**. In particular, the procedure **200** proceeds to stage **210** when the first material comprises a SAM material. In other embodiments, the procedure **200** proceeds to stage **212** and bypasses stage **210**.

In stage **210**, a third material is formed on the IC structure. The third material may be formed on the IC structure when the first material **502** comprises a SAM material. The third material may comprise a second SAM material that is different from a SAM material of the first material. In some embodiments, the second SAM material may comprise aminosilanes (such as Me2N—SiR3, (Me2N)2SiR2, (Me2N)3SiR, where R is long chain alkyls from C8H17 to C18H37), octadecylphosphonic acid, octadecylthiol, chlorosilane, alkoxysilane (with long alkane chains (such as octadecyl trichlorosilane, trimethoxy(octadecyl)silane)) or fluorocarbon (such as triethoxy(3,3,4,4,5,5,6,6,7,7,8,8,9,9,10,10,10-heptadecafluorodecyl), and 1-(3,3,4,4,5,5,6,6,7,7,8,8,9,9,10,10,10-heptadecafluorodecyl)-N,N,N',N',N",N"-hexamethylsilanetriamine) silane chains, hydrophobic polymer, HfO2, ZrO2, or some combination thereof. The third material is formed within the recess and attaches to the portions of the first dielectric material and/or the second dielectric material from which the first material was removed via the angled etch. The third material may attach to Si—OH groups produced on the sidewalls of the recess during the angled etch procedure.

FIG. **11** illustrates an IC structure **1100** according to stage **210** of the procedure **200** of FIG. **2**, according to some embodiments of the present disclosure. In particular, FIG. **11** illustrates a result of performing stage **210** on the IC structure **600** (FIG. **6**). It should be understood that stage **210** can be performed on the IC structure **700** (FIG. **7**) or the IC structure **800** (FIG. **8**), where the third material formed in stage **210** attaches to the portions of the first dielectric material **302** and/or the second dielectric material **304** from which the first material **502** was removed on the IC structure **700** or the IC structure **800**.

The IC structure **1100** includes third material **1102** located at the first sidewall **410** and the second sidewall **412**. In particular, a first portion **1104** of the third material **1102** is located at the first sidewall **410** and a second portion **1106** of the third material **1102** is located at the second sidewall **412**. The third material **1102** is located at the sidewalls where the first material **502** was removed in stage **208**. The third material **1102** defines the sidewalls of the recess **402** when formed on the first sidewall **410** and the second sidewall **412**. In particular, the third material **1102** redefines the sidewalls of the recess **402** as a first sidewall **1108** and a second sidewall **1110**. The first portion **1104** of the third material **1102** is located between the first portion **404** of the second dielectric material **304** and the recess **402**, and the second portion **1106** of the third material **1102** is located between the second portion **406** of the second dielectric material **304** and the recess **402**, where the first sidewall **1108** defined by the first portion **1104** and the second sidewall **1110** defined by the second portion **1106** abut the recess **402**. From stage **210**, the procedure **200** may proceed to stage **212**.

In stage **212**, a selective growth procedure is performed. The selective growth procedure includes a bottom-up growth of a second material on the first material. The second material can attach to the first material and itself without attaching to the sidewalls of the recess, thereby attaching to the first material and forming outwards from the first material. The second material can grow, through attaching to the first material and itself, to fill the remainder of the recess, where the growth of the second material can prevent seams and/or voids that can occur with other recess fill procedures. The second material can comprise a dielectric material (such as SiOx, SiN, SiC, SiCN, SiOC, SiOCN, and/or metal oxides including ZrO2, HfO2, and AlO2), a metal (such as Ti, TiN, Ta, TaN, Al, Cu, Co, Ru, and/or W), a sacrificial material that can be removed in a subsequent procedure (such as TiO2, TiN, SiOx, SiN, SiC, SiCN, SiOC, SiOCN, Ti, Ta, TaN, Al, Cu, Co, Ru, W, and/or metal oxides including ZrO2, HfO2, and AlO2), or some combination thereof.

FIG. **9** illustrates an IC structure **900** according to stage **212** of the procedure **200** of FIG. **2**, according to some embodiments of the present disclosure. In particular, FIG. **9** illustrates a result of performing stage **212** on the IC structure **600** (FIG. **6**) in some embodiments.

The IC structure **900** includes a second material **902** located within the recess **402** (FIG. **4**). The second material **902** is located on the first material **502** and abuts the first material **502**. The second material **902** may have been grown on the first material **502** while not growing on the second dielectric material **304**. The second material **902** further is located between the first portion **404** of the second dielectric material **304** and the second portion **406** of the second dielectric material **304**. The second material **902** further abuts the first portion **404** and the second portion **406**. In particular, the second material **902** is in contact with the first sidewall **410** defined by the first portion **404** of the second dielectric material **304** and is in contact with the second sidewall **412** defined by the second portion **406** of the second dielectric material **304**. The second material **902** can fill the remainder of the recess **402**, such that the first material **502** and the second material **902** collectively fill the recess **402**.

FIG. **12** illustrates another IC structure **1200** according to stage **212** of the procedure **200** of FIG. **2**, according to some embodiments of the present disclosure. In particular, FIG. **12** illustrates a result of performing stage **212** on the IC structure **1100** (FIG. **11**) in some embodiments.

The IC structure **1200** includes a second material **1202** located within the recess **402** (FIG. **4**). The second material **1202** is located on the first material **502** and abuts the first material **502**. The second material **1202** may have been grown on the first material **502** while not growing on the third material **1102**. In some embodiments, the third material **1102** may have been formed to prevent the second material **1202** from attaching to the sidewalls and a material comprising the third material **1102** may have been selected to be a material to which the second material **1202** will not attach when being grown. The second material **1202** is located between the first portion **1104** of the third material **1102** and the second portion **1106** of the third material **1102**, where the first portion **1104** is located between a first portion **404** of the second dielectric material **304** and the second material **1202**, and the second portion **1106** is located between the second portion **406** of the second dielectric material **304** and the second material **1202**. The second material **1202** abuts the first portion **1104** of the third material **1102** and the second portion **1106** of the third material **1102**. In particular, the second material **1202** abuts the first sidewall **1108** defined by the first portion **1104** of the third material **1102** and the second sidewall **1110** defined by the second portion **1106** of the third material **1102**. The second material **1202** can fill the remainder of the recess **402**, such that the first material **502**, the third material **1102**, and the second material **1202** collectively fill the recess. From stage **212**, the procedure **200** can proceed to stage **214**. In other embodiments, the procedure **200** may finish with the completion of stage **212**.

In stage **214**, the second material is removed. In particular, stage **214** can be performed when the second material is a sacrificial material. In some embodiments, the first material and/or the third material may be removed with the second material in stage **214**. Further procedures may be performed after stage **214** to fill the recess and/or cover the recess after the second material, the first material, and/or the third material have been removed. In some embodiments, stage **214** may be omitted.

FIG. **10** illustrates an IC structure **1000** according to stage **214** of the procedure **200** of FIG. **2**, according to some embodiments of the present disclosure. In particular, FIG. **10** illustrates a result of performing stage **214** on the IC structure **900** (FIG. **9**). In particular, the second material **902** (FIG. **9**) has been removed in the illustrated embodiment of the IC structure **1000**. It should be understood that stage **214** may be performed on the IC structure **1200** (FIG. **12**), where the second material **1202** (FIG. **12**), the third material **1102** (FIG. **11**), and/or the first material **502** (FIG. **5**) may be removed from the IC structure **1200**.

In the IC structure **1000** illustrated, the second material **902** had been removed. A void **1002** (i.e., lack of material) exists in the recess **402** where the second material **902** had been removed. The removal of the second material **902** exposes the first material **502**, the first sidewall **410**, and the second sidewall **412**. The void **1002** may be filled or covered via further procedures performed on the IC structure **1000**. In other embodiments, the first material **502**, or some portion of, may be removed with the second material **902**, where the void **1002** includes the area in the recess **402** where the second material **902** and the first material **502** were removed.

While the figures illustrated to describe the procedure **200** are illustrated utilizing the IC structure **600** (FIG. **6**) with the first material **502** being a uniform thickness, it is to be understood that the procedure **200** can produce IC structures with first material that varies in thickness. For example, the procedure **200** can be completed with the IC structure **700** (FIG. **7**) and IC structure **800** (FIG. **8**), where the first material **502** can have the shapes illustrated for IC structure **700** or IC structure **800** in other embodiments. Further, while the recess **402** (FIG. **4**) has vertical sidewalls in the illustrated embodiment, it is to be understood that the recess **402** can be different shapes in other embodiments. For example, the recess **402** may have tapered sidewalls, as illustrated in FIG. **22**, or reentrant sidewalls, as illustrated in FIG. **23**, in other embodiments.

FIG. **13** illustrates another procedure **1300** for producing IC structures, according to some embodiments of the present disclosure. FIG. **14** through FIG. **20** illustrates IC structures corresponding to different stages of the procedure **1300**.

The procedure **1300** can begin with a formed first dielectric material, such as the first dielectric material **1402** illustrated in FIG. **14**. The first dielectric material may include one or more of the features of the first dielectric material **102** (FIG. **1**), including the materials of which the first dielectric material **102** may be comprised. The procedure **1300** initiates with stage **1302**. In stage **1302**, a second dielectric material is applied to the first dielectric material. The second dielectric material may include one or more of the features of the second dielectric material **104** (FIG. **1**), including the materials of which the second dielectric material **104** may be comprised. In particular, the second dielectric material is applied to a top surface of the first dielectric material. The second dielectric material can be applied at approximately a uniform thickness across the top surface of the first dielectric material and can extend for an entirety of the top surface of the first dielectric material. The procedure **1300** may proceed from stage **1302** to stage **1304**.

FIG. **14** illustrates an IC structure **1400** according to stage **1302** of the procedure **1300** of FIG. **13**, according to some embodiments of the present disclosure. The IC structure **1400** includes a first dielectric material **1402**. The IC structure **1400** further includes a second dielectric material **1404** located on the first dielectric material **1402**. In particular, the second dielectric material **1404** is located on a top surface **1406** of the first dielectric material **1402**. The second dielectric material **1404** extends on the top surface **1406** of the first dielectric material **1402** for an entirety of the top surface **1406**, where the second dielectric material **1404** has an approximately uniform thickness for the entirety of the top surface **1406**.

In stage **1304**, patterning of the second dielectric material is performed. Patterning of the second dielectric material may include removing a portion of the second dielectric material from the top surface of the first dielectric material to produce a certain pattern. In some embodiments, patterning of the second dielectric material may include curing of a portion of the second dielectric material and removing the uncured portion of the second dielectric material or cured portion of the second dielectric material, depending on the chemical makeup of the second dielectric material and the removal procedure being applied. For the disclosed embodiments, the pattern produced by the patterning of the second dielectric material includes at least one recess. The patterned second dielectric material and the first dielectric material may collectively be referred to as a base assembly. The procedure **1300** may proceed from stage **1304** to stage **1306**.

FIG. **15** illustrates an IC structure **1500** according to stage **1304** of the procedure **1300** of FIG. **13**, according to some embodiments of the present disclosure. In particular, FIG. **15** illustrates the IC structure **1500** produced by stage **1304** being applied to the IC structure **1400** (FIG. **14**) in accordance with some embodiments disclosed herein.

The performance of stage **1304** caused a portion of the second dielectric material **1404** to be removed. The removal of the portion of the second dielectric material **1404** produces a recess **1502** within the second dielectric material **1404**, where the recess **1502** is located between a first portion **1504** of the second dielectric material **1404** and a second portion **1506** of the second dielectric material **1404**. In the illustrated embodiment, the first portion **1504** of the second dielectric material **1404** defines a first sidewall **1510** of the recess **1502** and the second portion **1506** of the second dielectric material **1404** defines a second sidewall **1512** of the recess **1502**. While the first sidewall **1510** and the second sidewall **1512** are illustrated as being straight and perpendicular to the top surface **1406** of the first dielectric material **1402** in the illustrated embodiment, it is to be understood that the first sidewall **1510** and/or the second sidewall **1512** can be curved, and/or tapered or reentrant in other embodiments. The recess **1502** extends from a top surface **1508** of the second dielectric material **1404** to the top surface **1406** of the first dielectric material **1402**, where the top surface **1508** of the second dielectric material **1404** includes a top surface of the first portion **1504** and the second portion **1506** of the second dielectric material **1404**. The top surface **1406** of the first dielectric material **1402** defines a bottom **1514** of the recess **1502**. In other embodiments, the recess **1502** may extend partially through the second dielectric material **1404**, or through the second dielectric material **1404** and partially through the first dielectric material **1402**.

In stage **1306**, a first material is formed on the IC structure. The first material may include one or more of the features of the first material **116** (FIG. **1**), including the materials of which the first material **116** may be comprised. In particular, the first material is formed on the top surface of the second dielectric material and within the recess at the sidewalls and the bottom of the recess. The first material may be deposited on the top surface of the second dielectric material and within the recess at the sidewalls and the bottom of the recess via a deposition procedure. The first material has a substantially uniform thickness from the surface to which it is applied (i.e., the top surface of the second dielectric material, the sidewalls of the recess, or the bottom of the recess). The procedure **1300** may proceed from stage **1306** to stage **1308**.

FIG. **16** illustrates an IC structure **1600** according to stage **1306** of the procedure **1300** of FIG. **13**, according to some embodiments of the present disclosure. In particular, FIG. **16** illustrates the IC structure **1600** produced by stage **1306** being applied to the IC structure **1500** (FIG. **15**) in accordance with some embodiments disclosed herein.

The IC structure **1600** includes a first material **1602** that has been formed on the top surface **1508** of the second dielectric material **1404**, at the first sidewall **1510** and the second sidewall **1512** within the recess **1502**, and at the bottom **1514** of the recess **1502**. The first material **1602** abuts the top surface **1508** of the second dielectric material **1404**, the first sidewall **1510**, the second sidewall **1512**, and the bottom **1514** of the recess **1502**. The first material **1602** has a substantially uniform thickness from the surfaces on which the first material **1602** has been formed.

In stage **1308**, a sacrificial material is to be formed on the first material. In particular, the sacrificial material may be formed on a portion of the first material that is not to be removed in stage **1310**. The sacrificial material can be formed on the first material located at the bottom of the recess and can prevent removal of the first material at the bottom of the recess during stage **1310**. The sacrificial material may comprise a material that will not be removed by an etch procedure that is performed in stage **1310**, but can be removed by another procedure, such as a chemical removal procedure and/or a curing removal procedure. The procedure **1300** may proceed from stage **1308** to stage **1310**.

FIG. **17** illustrates an IC structure **1700** according to stage **1308** of the procedure **1300** of FIG. **13**, according to some embodiments of the present disclosure. In particular, FIG. **17** illustrates the IC structure **1700** produced by stage **1308** being applied to the IC structure **1600** (FIG. **16**) in accordance with some embodiments disclosed herein.

The IC structure **1700** includes a sacrificial material **1702** that has been formed on the first material **1602**. In particular, the sacrificial material **1702** has been formed within the recess **1502** on top of a portion of the first material **1602** located at the bottom **1514** of the recess **1502**. Further, the sacrificial material **1702** extends between the first material **1602** located on the first sidewall **1510** and the first material **1602** located on the second sidewall **1512**. A top surface **1704** of the sacrificial material **1702** is located between the top surface **1508** of the second dielectric material **1404** and the first material **1602** located at the bottom **1514** of the recess **1502**. Accordingly, a portion of the first material **1602** located on the first sidewall **1510** and a portion of the first material **1602** located on the second sidewall **1512** may remain exposed with the sacrificial material **1702** located on the first material **1602** at the bottom **1514** of the recess **1502**. In some embodiments, a majority (i.e., greater than 50%) of the first material **1602** located on the first sidewall **1510** and a majority (i.e., greater than 50%) of the first material located on the second sidewall **1512** remain exposed.

In stage **1310**, a portion of the first material on the IC structure is removed. Removal of the first material can include performing an etch procedure on the IC structure, where the etch procedure removes the first material. The etch procedure may include a perpendicular etch procedure (where ions of the etch procedure are directed substantially perpendicularly the bottom of the recess) or an angled etch procedure (where ions of the etch procedure are directed at a non-perpendicular angle to the bottom of the recess). In some embodiments, the ions may be directed at one or more angles within a predetermined distribution of angles, where the distribution can be less than or equal to 90 degrees and greater than or equal to 60 degrees in some embodiments. The etch procedure may remove a portion of the first material formed on the IC structure in stage **1306** that is not protected from removal by the sacrificial material formed in stage **1310**. The etch procedure can cause the portion of the first material contacted by ions of the etch procedure to be removed from the IC structure. In some embodiments, the ions may comprise fluorine (F) ions, where the angled etch utilizes an F+-based chemistry. In other embodiments, the ions may comprise ions of other materials utilized for etching procedures, such as chlorine (CI) ions in a Cl+-based chemistry or bromine (Br) ions in a Br+-based chemistry. In some embodiments, traces of the ions may remain on the IC structure after the etch has been performed. A de-scum or de-ash procedure may be performed on the IC structure to remove the traces of the ions in some embodiments.

FIG. **18** illustrates an IC structure **1800** according to stage **1310** of the procedure **1300** of FIG. **13**, according to some embodiments of the present disclosure. In particular, FIG. **18** illustrates the IC structure **1800** produced by stage **1310** being applied to the IC structure **1700** (FIG. **17**) in accordance with some embodiments disclosed herein. The procedure **1300** may proceed from stage **1310** to stage **1312**.

FIG. **18** illustrates ions **1802** that are directed at the IC structure **1800**. Arrows **1804** illustrate directions of travel of the ions **1802**. The ions **1802** are directed at the top surface **1508** of the second dielectric material **1404** and are directed perpendicular to the bottom **1514** of the recess **1502**. In other embodiments, the ions **1802** may be directed at a non-perpendicular angle (such as an angle that is less than 90 degrees and more than 60 degrees) to the bottom **1514** of the recess **1502**. The ions **1802** can contact the first material **1602** not being blocked by the sacrificial material **1702**. For example, the ions **1802** can contact the first material **1602** located on the top surface **1508** of the second dielectric material **1404** and the first material **1602** located on the first sidewall **1510** and the second sidewall **1512** (collectively referred to as "the sidewalls"), while being prevented from contacting the first material **1602** located at the bottom **1514** of the recess **1502** by the sacrificial material **1702**. In some embodiments, the ions **1802** may further contact a first portion of the first material **1602** located at the bottom **1514** of the recess **1502**, while being prevented from contacting a second portion of the first material **1602** located at the bottom **1514** of the recess **1502** by the sacrificial material **1702**. The ions **1802** remove the first material **1602** that the ions **1802** contact. In the illustrated embodiment, a thickness of the first material **1602** remaining after the etch procedure is substantially uniform. In other embodiments, portions of the first material **1602** not located directly between the sacrificial material **1702** and the bottom **1514** of the recess **1502** may be thicker than the portion of the first material **1602** located directly between the sacrificial material **1702** and the bottom **1514** of the recess **1502**, may be thinner than the portion of the first material **1602** located directly between the sacrificial material **1702** and the bottom **1514** of the recess **1502**, or may be completely removed. The first material **1602** may be removed from portions of the first sidewall **1510** and the second sidewall **1512**, thereby exposing the portions of the first sidewall **1510** and the second sidewall **1512**. In some embodiments, a majority (i.e., greater than 50%) of the first sidewall **1510** is exposed and a majority (i.e., greater than 50%) of the second sidewall **1512** is exposed when the portion of the first material **1602** has been removed.

In stage **1312**, the sacrificial material may be removed. In particular, the sacrificial material may be removed from the first material, thereby exposing the first material. For example, the sacrificial material may be removed via a chemical removal procedure and/or a curing removal procedure.

FIG. **19** illustrates an IC structure **1900** according to stage **1312** of the procedure **1300** of FIG. **13**, according to some embodiments of the present disclosure. In particular, FIG. **19** illustrates the IC structure **1900** produced by stage **1312** being applied to the IC structure **1800** (FIG. **18**) in accordance with some embodiments disclosed herein.

The IC structure **1900** is illustrated with the sacrificial material **1702** (FIG. **17**) removed. In particular, the sacrificial material **1702** has been removed from the first material **1602**. Accordingly, an entirety of a top surface **1902** of the first material **1602** is exposed. The first material **1602** may maintain the shape that the first material **1602** was prior to the removal of the sacrificial material **1702**. For example, the first material **1602** may have a substantially uniform thickness in the illustrated embodiment.

From stage **1312**, the procedure **1300** may proceed to stage **1314** or stage **1316** depending on the chemical makeup of the first material applied in stage **1306**. In particular, the procedure **1300** proceeds to stage **1314** when the first material comprises a SAM material. In other embodiments, the procedure **1300** proceeds to stage **1316** and bypasses stage **1314**.

In stage **1314**, a third material is formed on the IC structure. The third material may be formed on the IC structure when the first material comprises a SAM material. The third material may comprise a second SAM material that is different from a SAM material of the first material. In some embodiments, the second SAM material may comprise aminosilanes (such as Me2N—SiR3, (Me2N)2SiR2, (Me2N)3SiR, where R is long chain alkyls from C8H17 to C18H37), octadecylphosphonic acid, octadecylthiol, chlorosilane, alkoxysilane (with long alkane chains (such as octadecyl trichlorosilane, trimethoxy(octadecyl)silane) or fluorocarbon (such as triethoxy(3,3,4,4,5,5,6,6,7,7,8,8,9,9,10,10,10-heptadecafluorodecyl, and 1-(3,3,4,4,5,5,6,6,7,7,8,8,9,9,10,10,10-heptadecafluorodecyl)-N,N,N',N',N",N"-hexamethylsilanetriamine) silane chains, hydrophobic polymer, HfO2, ZrO2, or some combination thereof. The third material is formed within the recess and attaches to the portions of the first dielectric material and/or the second dielectric material from which the first material was removed via the etch. The third material may attach to Si—OH groups produced on the sidewalls of the recess during the etch procedure. Stage **1314** may produce the IC structure **1100** (FIG. **11**), where the first dielectric material

1402 (FIG. **14**) corresponds to the first dielectric material **302** (FIG. **3**), the second dielectric material **1404** (FIG. **14**) corresponds to the second dielectric material **304** (FIG. **3**), and the first material **1602** (FIG. **16**) corresponds to the first material **502** (FIG. **5**), and where each of the materials include the features of the corresponding materials of FIG. **11**. Further, the third material formed in stage **1314** includes the features of the third material **1102** (FIG. **11**) and can be formed on the portions of the sidewalls and/or the bottom **1514** of the recess **1502** from which the first material **1602** was removed in stage **1310**.

In stage **1316**, a selective growth procedure is performed. The selective growth procedure includes a bottom-up growth of a second material on the first material. The second material can attach to the first material and itself without attaching to the sidewalls of the recess, thereby attaching to the first material and forming outwards from the first material. The second material can grow, through attaching to the first material and itself, to fill the remainder of the recess, where the growth of the second material can prevent seams and/or voids that can occur with other recess fill procedures. The second material can comprise a dielectric material (such as SiOx, SiN, SiC, SiCN, SiOC, SiOCN, and/or metal oxides including ZrO2, HfO2, and AlO2), a metal (such as Ti, TiN, Ta, TaN, Al, Cu, Co, Ru, and/or W), a sacrificial material that can be removed in a subsequent procedure (such as TiO2, TiN, SiOx, SiN, SiC, SiCN, SiOC, SiOCN, Ti, Ta, TaN, Al, Cu, Co, Ru, W, and/or metal oxides including ZrO2, HfO2, and AlO2), or some combination thereof.

FIG. **20** illustrates an IC structure **3000** according to stage **1316** of the procedure **1300** of FIG. **13**, according to some embodiments of the present disclosure. In particular, FIG. **20** illustrates a result of performing stage **1316** on the IC structure **1900** (FIG. **19**) in some embodiments.

The IC structure **3000** includes a second material **3002** located within the recess **1502** (FIG. **15**). The second material **3002** is located on the first material **1602** and abuts the first material **1602**. The second material **3002** may have been grown on the first material **1602** while not growing on the second dielectric material **1404**. The second material **3002** further is located between the first portion **1504** of the second dielectric material **1404** and the second portion **1506** of the second dielectric material **1404**. The second material **3002** further abuts the first portion **1504** and the second portion **1506**. In particular, the second material **3002** is in contact with the first sidewall **1510** defined by the first portion **1504** of the second dielectric material **1404** and is in contact with the second sidewall **1512** defined by the second portion **1506** of the second dielectric material **1404**. The second material **3002** can fill the remainder of the recess **1502**, such that the first material **1602** and the second material **3002** collectively fill the recess **1502**.

It should be understood that stage **1316** can be applied to the IC structure **1100** (FIG. **11**), produced via stage **1314**, in other embodiments. In particular, the selective growth procedure can produce the second material within the recess of the IC structure. Applying stage **1316** to the IC structure **1100** can result in the IC structure **1200** (FIG. **12**), where the second material formed via stage **1316** corresponds to the second material **1202** (FIG. **12**), and the materials of the IC structure produced by stage **1316** include the features of the corresponding materials of the IC structure **1200**.

In stage **1318**, the second material is removed. In particular, stage **1318** can be performed when the second material is a sacrificial material. In some embodiments, the first material and/or the third material may be removed with the second material in stage **1318**. Further procedures may be performed after stage **1318** to fill the recess and/or cover the recess after the second material, the first material, and/or the third material have been removed.

FIG. **21** illustrates an IC structure **3100** according to stage **1318** of the procedure **1300** of FIG. **13**, according to some embodiments of the present disclosure. In particular, FIG. **21** illustrates a result of performing stage **1318** on the IC structure **3000** (FIG. **20**). In particular, the second material **3002** (FIG. **20**) has been removed in the illustrated embodiment of the IC structure **3100**. It should be understood that stage **1318** may be performed on the IC structure **1200** (FIG. **12**) produced by stage **1316**, where the second material, the third material, and the first material **502** may be removed from the IC structure **1200**.

In the IC structure **3100** illustrated, the second material **3002** had been removed. A void **3102** (i.e., lack of material) exists in the recess **1502** where the second material **3002** had been removed. The removal of the second material **3002** exposes the first material **1602**, the first sidewall **1510**, and the second sidewall **1512**. The void **3102** may be filled or covered via further procedures performed on the IC structure **3100**. In other embodiments, the first material **1602**, or some portion of, may be removed with the second material **3002**, where the void **3102** includes the area in the recess **1502** where the second material **3002** and the first material **1602** were removed.

While the figures illustrated to describe the procedure **1300** are illustrated utilizing the IC structure **1900** (FIG. **19**) with the first material **1602** being a uniform thickness, it is to be understood that the procedure **1300** can produce IC structures with first material that varies in thickness. For example, the thickness of a portion of the first material near the sidewalls of the recess may be thicker or thinner than another portion of the first material near the center of the recess in other embodiments. Further, while the recess **1502** (FIG. **5**) has vertical sidewalls in the illustrated embodiment, it is to be understood that the recess **1502** can be different shapes in other embodiments. For example, the recess **1502** may have tapered sidewalls, as illustrated in FIG. **22**, or reentrant sidewalls, as illustrated in FIG. **23**, in other embodiments.

FIG. **22** illustrates another IC structure **3200**, according to some embodiments of the present disclosure. In particular, the IC structure **3200** includes tapered sidewalls. Further, it is to be understood that the procedure **200** (FIG. **2**) and/or the procedure **1300** (FIG. **13**) can produce IC structures with tapered sidewalls. The IC structure **3200** may undergo further processing in IC die production where additional materials can be affixed on the top and/or bottom of the IC structure **3200**. A further IC component may be coupled to the IC die with the IC structure **3200** in some embodiments.

The IC structure **3200** may include one or more of the features of the IC structure **100** (FIG. **1**). In particular, the IC structure **3200** includes a first dielectric material **3202** that includes one or more features of the first dielectric material **102** (FIG. **1**), a second dielectric material **3204** that includes one or more of the features of the second dielectric material **104** (FIG. **1**), a first material **3206** that includes one or more features of the first material **116** (FIG. **1**), and a second material **3208** that includes one or more of the features of the second dielectric material **104** (FIG. **1**). The IC structure **3200** includes a recess **3210** located on the first dielectric material **3202** and between a first portion **3212** and a second portion **3214** of the second dielectric material **3204**, where the recess **3210** is filled by the first material **3206** and the second material **3208**, the second material **3208** being located on the first material **3206**. The recess **3210** is narrower at a top surface **3216** of the first dielectric material **3202** than at a top surface **3218** of the second dielectric material **3204**. Accordingly, a first sidewall **3220** of the recess **3210** and a second sidewall **3222** are both tapered from the top surface **3218** of the second dielectric material **3204** to the top surface **3216** of the first dielectric material **3202**. It is to be understood that any of the IC structures disclosed herein may have tapered sidewalls, such as the first sidewall **3220** and the second sidewall **3222**.

FIG. **23** illustrates another IC structure **3300**, according to some embodiments of the present disclosure. In particular, the IC structure **3300** includes reentrant sidewalls. Further, it is to be understood that the procedure **200** (FIG. **2**) and/or the procedure **1300** (FIG. **13**) can produce IC structures with reentrant sidewalls. The IC structure **3300** may undergo further processing in IC die production where additional materials can be affixed on the top and/or bottom of the IC structure **3300**. A further IC component may be coupled to the IC die with the IC structure **3300** in some embodiments.

The IC structure **3300** may include one or more of the features of the IC structure **100** (FIG. **1**). In particular, the IC structure **3300** includes a first dielectric material **3302** that includes one or more features of the first dielectric material **102** (FIG. **1**), a second dielectric material **3304** that includes one or more of the features of the second dielectric material **104** (FIG. **1**), a first material **3306** that includes one or more features of the first material **116** (FIG. **1**), and a second material **3308** that includes one or more of the features of the second dielectric material **104** (FIG. **1**). The IC structure **3300** includes a recess **3310** located on the first dielectric material **3302** and between a first portion **3312** and a second portion **3314** of the second dielectric material **3304**, where the recess **3310** is filled by the first material **3306** and the second material **3308**, the second material **3308** being located on the first material **3306**. The recess **3310** is wider at a top surface **3316** of the first dielectric material **3302** than at a top surface **3318** of the second dielectric material **3304**. Accordingly, a first sidewall **3320** of the recess **3310** and a second sidewall **3322** are both reentrant from the top surface **3318** of the second dielectric material **3304** to the top surface **3316** of the first dielectric material **3302**. It is to be understood that any of the IC structures disclosed herein may have reentrant sidewalls, such as the first sidewall **3320** and the second sidewall **3322**.

Example Structures and Devices

IC structures that include one or more recesses processed via selective growth as disclosed herein may be included in any suitable electronic device. FIGS. **24-28** illustrate various examples of devices and components that may include one or more recesses processed via selective grows as disclosed herein.

FIGS. **24**A-**24**B are top views of a wafer **2000** and dies **2002** that may include one or more IC structures having recesses processed via selective growth in accordance with any of the embodiments disclosed herein. In some embodiments, the dies **2002** may be included in an IC package, in accordance with any of the embodiments disclosed herein. For example, any of the dies **2002** may serve as any of the dies **2256** in an IC package **2200** shown in FIG. **25**. The wafer **2000** may be composed of semiconductor material and may include one or more dies **2002** having IC structures formed on a surface of the wafer **2000**. Each of the dies **2002** may be a repeating unit of a semiconductor product that includes any suitable IC. After the fabrication of the semiconductor product is complete (e.g., after manufacture of any embodiment of the IC structure **100** described herein), the wafer **2000** may undergo a singulation process in which each of the dies **2002** is separated from one another to provide discrete "chips" of the semiconductor product. In particular, devices that include one or more IC structures as disclosed herein may take the form of the wafer **2000** (e.g., not singulated) or the form of the die **2002** (e.g., singulated). The die **2002** may include one or more IC structures having recesses processed via selective growth, as well as, optionally, supporting circuitry to route electrical signals via the IC structures, as well as to any other IC components. In some embodiments, the wafer **2000** or the die **2002** may implement an RF FE device, a memory device (e.g., a static random-access memory (SRAM) device), a logic device (e.g., an AND, OR, NAND, or NOR gate), or any other suitable circuit element. Multiple ones of these devices may be combined on a single die **2002**.

FIG. **25** is a side, cross-sectional view of an example IC package **2200** that may include one or more IC structures having one or more recesses processed via selective growth in accordance with any of the embodiments disclosed herein. In some embodiments, the IC package **2200** may be a system-in-package (SiP).

As shown in FIG. **25**, the IC package **2200** may include a package substrate **2252**. The package substrate **2252** may be formed of a dielectric material (e.g., a ceramic, a glass, a combination of organic and inorganic materials, a buildup film, an epoxy film having filler particles therein, etc., and may have embedded portions having different materials), and may have conductive pathways extending through the dielectric material between the face **2272** and the face **2274**, or between different locations on the face **2272**, and/or between different locations on the face **2274**.

The package substrate **2252** may include conductive contacts **2263** that are coupled to conductive pathways **2262** through the package substrate **2252**, allowing circuitry within the dies **2256** and/or the interposer **2257** to electrically couple to various ones of the conductive contacts **2264** (or to other devices included in the package substrate **2252**, not shown).

The IC package **2200** may include an interposer **2257** coupled to the package substrate **2252** via conductive contacts **2261** of the interposer **2257**, first-level interconnects **2265**, and the conductive contacts **2263** of the package substrate **2252**. The first-level interconnects **2265** illustrated in FIG. **25** are solder bumps, but any suitable first-level interconnects **2265** may be used. In some embodiments, no interposer **2257** may be included in the IC package **2200**; instead, the dies **2256** may be coupled directly to the conductive contacts **2263** at the face **2272** by first-level interconnects **2265**.

The IC package **2200** may include one or more dies **2256** coupled to the interposer **2257** via conductive contacts **2254** of the dies **2256**, first-level interconnects **2258**, and conductive contacts **2260** of the interposer **2257**. The conductive contacts **2260** may be coupled to conductive pathways (not shown) through the interposer **2257**, allowing circuitry within the dies **2256** to electrically couple to various ones of the conductive contacts **2261** (or to other devices included in the interposer **2257**, not shown). The first-level interconnects **2258** illustrated in FIG. **25** are solder bumps, but any suitable first-level interconnects **2258** may be used. As used herein, a "conductive contact" may refer to a portion of electrically conductive material (e.g., metal) serving as an interface between different components; conductive contacts may be recessed in, flush with, or extending away from a surface of a component, and may take any suitable form (e.g., a conductive pad or socket).

In some embodiments, an underfill material **2266** may be disposed between the package substrate **2252** and the interposer **2257** around the first-level interconnects **2265**, and a mold compound **2268** may be disposed around the dies **2256** and the interposer **2257** and in contact with the package substrate **2252**. In some embodiments, the underfill material **2266** may be the same as the mold compound **2268**. Example materials that may be used for the underfill material **2266** and the mold compound **2268** are epoxy mold materials, as suitable. Second-level interconnects **2270** may be coupled to the conductive contacts **2264**. The second-level interconnects **2270** illustrated in FIG. **25** are solder balls (e.g., for a ball grid array arrangement), but any suitable second-level interconnects **22770** may be used (e.g., pins in a pin grid array arrangement or lands in a land grid array arrangement). The second-level interconnects **2270** may be used to couple the IC package **2200** to another component, such as a circuit board (e.g., a motherboard), an interposer, or another IC package, as known in the art and as discussed below with reference to FIG. **26**.

The dies **2256** may take the form of any of the embodiments of the die **2002** discussed herein and may include any of the embodiments of an IC structure having one or more recesses processed via selective growth, e.g., any of the IC structures, described herein. In embodiments in which the IC package **2200** includes multiple dies **2256**, the IC package **2200** may be referred to as an MCP. The dies **2256** may include circuitry to perform any desired functionality. For example, one or more of the dies **2256** may be RF FE dies, including one or more recesses processed via selective growth as described herein, one or more of the dies **2256** may be logic dies (e.g., silicon-based dies), one or more of the dies **2256** may be memory dies (e.g., high bandwidth memory), etc. In some embodiments, any of the dies **2256** may include one or more recesses processed via selective growth, e.g., as discussed above; in some embodiments, at least some of the dies **2256** may not include any recesses processed via selective growth.

The IC package **2200** illustrated in FIG. **25** may be a flip chip package, although other package architectures may be used. For example, the IC package **2200** may be a ball grid array (BGA) package, such as an embedded wafer-level ball grid array (eWLB) package. In another example, the IC package **2200** may be a wafer-level chip scale package (WLCSP) or a panel fan-out (FO) package. Although two dies **2256** are illustrated in the IC package **2200** of FIG. **25**, an IC package **2200** may include any desired number of the dies **2256**. An IC package **2200** may include additional passive components, such as surface-mount resistors, capacitors, and inductors disposed on the first face **2272** or the second face **2274** of the package substrate **2252**, or on either face of the interposer **2257**. More generally, an IC package **2200** may include any other active or passive components known in the art.

FIG. **26** is a cross-sectional side view of an IC device assembly **2300** that may include components having one or more IC structures implementing one or more recesses processed via selective growth in accordance with any of the embodiments disclosed herein. The IC device assembly **2300** includes a number of components disposed on a circuit board **2302** (which may be, e.g., a motherboard). The IC device assembly **2300** includes components disposed on a first face **2340** of the circuit board **2302** and an opposing second face **2342** of the circuit board **2302**; generally, components may be disposed on one or both faces **2340** and **2342**. In particular, any suitable ones of the components of the IC device assembly **2300** may include any of the IC structures implementing one or more recesses processed via selective growth in accordance with any of the embodiments disclosed herein; e.g., any of the IC packages discussed below with reference to the IC device assembly **2300** may take the form of any of the embodiments of the IC package **2200** discussed above with reference to FIG. **25**.

In some embodiments, the circuit board **2302** may be a printed circuit board (PCB) including multiple metal layers separated from one another by layers of dielectric material and interconnected by electrically conductive vias. Any one or more of the metal layers may be formed in a desired circuit pattern to route electrical signals (optionally in conjunction with other metal layers) between the components coupled to the circuit board **2302**. In other embodiments, the circuit board **2302** may be a non-PCB substrate.

The IC device assembly **2300** illustrated in FIG. **26** includes a package-on-interposer structure **2336** coupled to the first face **2340** of the circuit board **2302** by coupling components **2316**. The coupling components **2316** may electrically and mechanically couple the package-on-interposer structure **2336** to the circuit board **2302**, and may include solder balls (e.g., as shown in FIG. **26**), male and female portions of a socket, an adhesive, an underfill material, and/or any other suitable electrical and/or mechanical coupling structure.

The package-on-interposer structure **2336** may include an IC package **2320** coupled to an interposer **2304** by coupling components **2318**. The coupling components **2318** may take any suitable form for the application, such as the forms discussed above with reference to the coupling components **2316**. The IC package **2320** may be or include, for example, a die (the die **2002** of FIG. **24**B), an IC device (e.g., the IC structure of FIGS. **1**, **3-12**, and **14-23**), or any other suitable component. In particular, the IC package **2320** may include one or more recesses processed via selective growth as described herein. Although a single IC package **2320** is shown in FIG. **26**, multiple IC packages may be coupled to the interposer **2304**; indeed, additional interposers may be coupled to the interposer **2304**. The interposer **2304** may provide an intervening substrate used to bridge the circuit board **2302** and the IC package **2320**. Generally, the interposer **2304** may spread a connection to a wider pitch or reroute a connection to a different connection. For example, the interposer **2304** may couple the IC package **2320** (e.g., a die) to a BGA of the coupling components **2316** for coupling to the circuit board **2302**. In the embodiment illustrated in FIG. **26**, the IC package **2320** and the circuit board **2302** are attached to opposing sides of the interposer **2304**; in other embodiments, the IC package **2320** and the circuit board **2302** may be attached to a same side of the interposer **2304**. In some embodiments, three or more components may be interconnected by way of the interposer **2304**.

The interposer **2304** may be formed of an epoxy resin, a fiberglass-reinforced epoxy resin, a ceramic material, or a polymer material such as polyimide. In some implementations, the interposer **2304** may be formed of alternate rigid or flexible materials that may include materials for use in a semiconductor substrate, such as silicon, germanium, and other group III-V and group IV materials. The interposer **2304** may include metal interconnects **2308** and vias **2310**, including but not limited to through-silicon vias (TSVs) **2306**. The interposer **2304** may further include embedded devices **2314**, including both passive and active devices. Such devices may include, but are not limited to, capacitors, decoupling capacitors, resistors, inductors, fuses, diodes, transformers, sensors, electrostatic discharge (ESD) protection devices, and memory devices. More complex devices such as further RF devices, power amplifiers, power management devices, antennas, arrays, sensors, and microelectromechanical systems (MEMS) devices may also be formed on the interposer **2304**. In some embodiments, the IC structures implementing one or more recesses processed via selective growth as described herein may also be implemented in/on the interposer **2304**. The package-on-interposer structure **2336** may take the form of any of the package-on-interposer structures known in the art.

The IC device assembly **2300** may include an IC package **2324** coupled to the first face **2340** of the circuit board **2302** by coupling components **2322**. The coupling components **2322** may take the form of any of the embodiments discussed above with reference to the coupling components **2316**, and the IC package **2324** may take the form of any of the embodiments discussed above with reference to the IC package **2320**.

The IC device assembly **2300** illustrated in FIG. **26** includes a package-on-package structure **2334** coupled to the second face **2342** of the circuit board **2302** by coupling components **2328**. The package-on-package structure **2334** may include an IC package **2326** and an IC package **2332** coupled together by coupling components **2330** such that the IC package **2326** is disposed between the circuit board **2302** and the IC package **2332**. The coupling components **2328** and **2330** may take the form of any of the embodiments of the coupling components **2316** discussed above, and the IC packages **2326** and **2332** may take the form of any of the embodiments of the IC package **2320** discussed above. The package-on-package structure **2334** may be configured in accordance with any of the package-on-package structures known in the art.

FIG. **27** is a block diagram of an example computing device **2400** that may include one or more components with one or more IC structures having one or more recesses processed via selective growth in accordance with any of the embodiments disclosed herein. For example, any suitable ones of the components of the computing device **2400** may include a die (e.g., the die **2002** (FIG. **24**B)) including one or more recesses processed via selective growth in accordance with any of the embodiments disclosed herein. Any of the components of the computing device **2400** may include an IC device (e.g., any embodiment of the IC structures of FIGS. **1**, **3**-**12**, and **14**-**23**) and/or an IC package **2200** (FIG. **25**). Any of the components of the computing device **2400** may include an IC device assembly **2300** (FIG. **26**).

A number of components are illustrated in FIG. **27** as included in the computing device **2400**, but any one or more of these components may be omitted or duplicated, as suitable for the application. In some embodiments, some or all of the components included in the computing device **2400** may be attached to one or more motherboards. In some embodiments, some or all of these components are fabricated onto a single SoC die.

Additionally, in various embodiments, the computing device **2400** may not include one or more of the components illustrated in FIG. **27**, but the computing device **2400** may include interface circuitry for coupling to the one or more components. For example, the computing device **2400** may not include a display device **2406**, but may include display device interface circuitry (e.g., a connector and driver circuitry) to which a display device **2406** may be coupled. In another set of examples, the computing device **2400** may not include an audio input device **2418** or an audio output device **2408**, but may include audio input or output device interface circuitry (e.g., connectors and supporting circuitry) to which an audio input device **2418** or audio output device **2408** may be coupled.

The computing device **2400** may include a processing device **2402** (e.g., one or more processing devices). As used herein, the term "processing device" or "processor" may refer to any device or portion of a device that processes electronic data from registers and/or memory to transform that electronic data into other electronic data that may be stored in registers and/or memory. The processing device **2402** may include one or more digital signal processors (DSPs), application-specific ICs (ASICs), central processing units (CPUs), graphics processing units (GPUs), cryptoprocessors (specialized processors that execute cryptographic algorithms within hardware), server processors, or any other suitable processing devices. The computing device **2400** may include a memory **2404**, which may itself include one or more memory devices such as volatile memory (e.g., DRAM), nonvolatile memory (e.g., read-only memory (ROM)), flash memory, solid-state memory, and/or a hard drive. In some embodiments, the memory **2404** may include memory that shares a die with the processing device **2402**. This memory may be used as cache memory and may include, e.g., eDRAM, and/or spin transfer torque magnetic random-access memory (STT-MRAM).

In some embodiments, the computing device **2400** may include a communication chip **2412** (e.g., one or more communication chips). For example, the communication chip **2412** may be configured for managing wireless communications for the transfer of data to and from the computing device **2400**. The term "wireless" and its derivatives may be used to describe circuits, devices, systems, methods, techniques, communications channels, etc., that may communicate data through the use of modulated electromagnetic radiation through a nonsolid medium. The term does not imply that the associated devices do not contain any wires, although in some embodiments they might not.

The communication chip **2412** may implement any of a number of wireless standards or protocols, including but not limited to Institute for Electrical and Electronic Engineers (IEEE) standards including Wi-Fi (IEEE 802.11 family), IEEE 802.16 standards (e.g., IEEE 802.16-2005 Amendment), Long-Term Evolution (LTE) project along with any amendments, updates, and/or revisions (e.g., advanced LTE project, ultramobile broadband (UMB) project (also referred to as "3GPP2"), etc.). IEEE 802.16 compatible Broadband Wireless Access (BWA) networks are generally referred to as WiMAX networks, an acronym that stands for Worldwide Interoperability for Microwave Access, which is a certification mark for products that pass conformity and interoperability tests for the IEEE 802.16 standards. The communication chip **2412** may operate in accordance with a Global System for Mobile Communication (GSM), General Packet Radio Service (GPRS), Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Evolved HSPA (E-HSPA), or LTE network. The communication chip **2412** may operate in accordance with Enhanced Data for GSM Evolution (EDGE), GSM EDGE Radio Access Network (GERAN), Universal Terrestrial Radio Access Network (UTRAN), or Evolved UTRAN (E-UTRAN). The communication chip **2412** may operate in accordance with Code Division Multiple Access (CDMA), Time Division Multiple Access (TDMA), Digital Enhanced Cordless Telecommunications (DECT), Evolution-Data Optimized (EV-DO), and derivatives thereof, as well as any other wireless protocols that are designated as 3G, 4G, 5G, and beyond. The communication chip **2412** may operate in accordance with other wireless protocols in other embodiments. The computing device **2400** may include an antenna **2422** to facilitate wireless communications and/or to receive other wireless communications (such as AM or FM radio transmissions).

In some embodiments, the communication chip **2412** may manage wired communications, such as electrical, optical, or any other suitable communication protocols (e.g., the Ethernet). As noted above, the communication chip **2412** may include multiple communication chips. For instance, a first communication chip **2412** may be dedicated to shorter-range wireless communications such as Wi-Fi or Bluetooth, and a second communication chip **2412** may be dedicated to longer-range wireless communications such as global positioning system (GPS), EDGE, GPRS, CDMA, WiMAX, LTE, EV-DO, or others. In some embodiments, a first communication chip **2412** may be dedicated to wireless communications, and a second communication chip **2412** may be dedicated to wired communications.

In various embodiments, IC structures including one or more recesses processed via selective growth as described herein may be particularly advantageous for use within the one or more communication chips **2412**, described above. For example, such IC structures may be used to implement one or more of power amplifiers, low-noise amplifiers, filters (including arrays of filters and filter banks), switches, upconverters, downconverters, and duplexers.

The computing device **2400** may include battery/power circuitry **2414**. The battery/power circuitry **2414** may include one or more energy storage devices (e.g., batteries or capacitors) and/or circuitry for coupling components of the computing device **2400** to an energy source separate from the computing device **2400** (e.g., AC line power).

The computing device **2400** may include a display device **2406** (or corresponding interface circuitry, as discussed above). The display device **2406** may include any visual indicators, such as a heads-up display, a computer monitor, a projector, a touchscreen display, a liquid crystal display (LCD), a light-emitting diode display, or a flat panel display, for example.

The computing device **2400** may include an audio output device **2408** (or corresponding interface circuitry, as discussed above). The audio output device **2408** may include any device that generates an audible indicator, such as speakers, headsets, or earbuds, for example.

The computing device **2400** may include an audio input device **2418** (or corresponding interface circuitry, as discussed above). The audio input device **2418** may include any device that generates a signal representative of a sound, such as microphones, microphone arrays, or digital instruments (e.g., instruments having a musical instrument digital interface (MIDI) output).

The computing device **2400** may include a GPS device **2416** (or corresponding interface circuitry, as discussed above). The GPS device **2416** may be in communication with a satellite-based system and may receive a location of the computing device **2400**, as known in the art.

The computing device **2400** may include an other output device **2410** (or corresponding interface circuitry, as discussed above). Examples of the other output device **2410** may include an audio codec, a video codec, a printer, a wired or wireless transmitter for providing information to other devices, or an additional storage device.

The computing device **2400** may include an other input device **2420** (or corresponding interface circuitry, as discussed above). Examples of the other input device **2420** may include an accelerometer, a gyroscope, a compass, an image capture device, a keyboard, a cursor control device such as a mouse, a stylus, a touchpad, a bar code reader, a Quick Response (QR) code reader, any sensor, or a radio frequency identification (RFID) reader.

The computing device **2400** may have any desired form factor, such as a handheld or mobile computing device (e.g., a cell phone, a smart phone, a mobile internet device, a music player, a tablet computer, a laptop computer, a netbook computer, an ultrabook computer, a personal digital assistant (PDA), an ultramobile personal computer, etc.), a desktop computing device, a server or other networked computing component, a printer, a scanner, a monitor, a set-top box, an entertainment control unit, a vehicle control unit, a digital camera, a digital video recorder, or a wearable computing device. In some embodiments, the computing device **2400** may be any other electronic device that processes data.

SELECT EXAMPLES

The following paragraphs provide various examples of the embodiments disclosed herein.

Example 1 includes an integrated circuit (IC) structure, comprising a first dielectric material, a second dielectric material on the first dielectric material, a recess in the second dielectric material, wherein the recess includes a bottom, a top, and sidewalls, a first material within the recess and at a bottom of the recess, wherein the first material includes a metal and oxygen, a self-assembled monolayer (SAM) material, or an organic material, and a second material within the recess and between the first material and the top of the recess, wherein the second material is in contact with the sidewalls of the recess.

Example 2 includes the IC structure of example 1, wherein a top edge of the second material is aligned with the top of the recess.

Example 3 includes the IC structure of example 1, wherein the first material is in contact with a first portion of the sidewalls of the recess, and wherein the second material is in contact with a second portion of the sidewalls of the recess.

Example 4 includes the IC structure of example 1, wherein the second material is in contact with an entirety of the sidewalls of the recess.

Example 5 includes the IC structure of example 1, wherein the second material comprises a dielectric material or a metal.

Example 6 includes the IC structure of example 1, further comprising a third material on both sides of the recess, wherein the third material is between the second dielectric material and the recess, and wherein the third material defines the sidewalls of the recess.

Example 7 includes the IC structure of example 1, wherein the bottom of the recess abuts the first dielectric material, and wherein the first material is on the first dielectric material.

Example 8 includes the IC structure of example 1, wherein the first material comprises silicon and oxide.

Example 9 includes an integrated circuit (IC) package, comprising an IC die, including a first dielectric material, a second dielectric material on the first dielectric material, wherein a recess is between a first portion of the second dielectric material and a second portion of the second dielectric material, and wherein a sidewall of the first portion of the second dielectric material abuts the recess and a sidewall of the second portion of the second dielectric material abuts the recess, and a material on the first dielectric material and within the recess, the material comprising a metal and oxygen, a self-assembled monolayer (SAM) material, or an organic material, wherein less than an entirety of the sidewall of the first portion of the second dielectric material abuts the material and less than an entirety of the sidewall of the second portion of the second dielectric material abuts the material, and a further IC component coupled to the IC die.

Example 10 includes the IC package of example 9, wherein a height of the material is less than a height of the recess.

Example 11 includes the IC package of example 9, wherein the material is a first material, wherein the IC die further includes a second material on the first material and within the recess, and wherein the second material and the first material fill the recess.

Example 12 includes the IC package of example 11, wherein the second material abuts a portion of the sidewall of the first portion of the second dielectric material and a portion of the sidewall of the second portion of the second dielectric material.

Example 13 includes the IC package of example 12, wherein the portion of the sidewall of the first portion of the second dielectric material is a first portion of the sidewall of the first portion of the second dielectric material, wherein the portion of the sidewall of the second portion of the second dielectric material is a first portion of the sidewall of the second portion of the second dielectric material, and wherein the first material abuts a second portion of the sidewall of the first portion of the second dielectric material and a second portion of the sidewall of the second portion of the second dielectric material.

Example 14 includes the IC package of example 9, wherein the material is a first material, and wherein the IC die further includes a second material on the first material and within the recess, and a third material, a first portion of the third material between the second material and the first portion of the second dielectric material and a second portion of the third material between the second material and the second portion of the second dielectric material.

Example 15 includes the IC package of example 14, wherein the SAM material is a first SAM material, wherein the first material comprises the first SAM material or an organic material, and wherein the third material comprises a second SAM material, the second SAM material being different from the first SAM material.

Example 16 includes the IC package of example 14, wherein the first material, the second material, and the third material fill the recess.

Example 17 includes a method of producing an integrated circuit (IC) structure, comprising forming a first material on a base assembly, the base assembly having a recess and the first material being applied to sidewalls and a bottom of the recess, performing an angled etch of the first material to remove the first material from a portion of the sidewalls of the recess while leaving the first material in place at the bottom of the recess, and forming a second material on the first material.

Example 18 includes the method of example 17, wherein performing the angled etch includes directing ions at an angle of between 60 degrees and 85 degrees to the bottom of the recess at the first material, the ions to remove the first material from the portion of the sidewalls of the recess.

Example 19 includes the method of example 17, wherein the base assembly comprises one or more dielectric materials, and wherein the first material comprises a metal and oxygen, a self-assembled monolayer (SAM) material, or an organic material.

Example 20 includes the method of example 19, wherein the SAM material is a first SAM material or an organic material, and wherein the method further comprises forming a third material on the portion of the sidewalls, the third material comprising a second SAM material.

Example 21 includes an integrated circuit (IC) structure, comprising a first dielectric material region, a second dielectric material region located on the first dielectric material region, a top edge of the second dielectric material region being opposite from the first dielectric material region, and a material region located on the first dielectric material region and between a first portion of the second dielectric material region and a second portion of the second dielectric material region, a top edge of the material region being located between the first dielectric material region and the top edge of the second dielectric material region, and the material region comprising a metal and oxygen, a self-assembled monolayer (SAM) material, or an organic material.

Example 22 includes the IC structure of example 21, further comprising a growth region located on the growth base region, the growth base region being located between the base structure and the growth region, wherein a top edge of the growth region is aligned with the top edge of the pattern region.

Example 23 includes the IC structure of example 22, wherein the growth region abuts a sidewall of the first portion of the pattern region and a sidewall of the second portion of the pattern region.

Example 24 includes the IC structure of example 22, wherein the growth region comprises a dielectric material or a metal.

Example 25 includes the IC structure of example 22, further comprising a protective region, a first portion of the protective region located between the first portion of the pattern region and the growth region and a second portion of the protective region located between the second portion of the pattern region, wherein the protective region and the growth base region separate the growth region from the base structure and the pattern region.

Example 26 includes the IC structure of example 25, wherein the growth base region comprises a SAM material or an organic material.

Example 27 includes the IC structure of example 21, wherein the growth base region includes silicon and oxide.

Example 28 includes the IC structure of example 21, wherein the base structure comprises a first dielectric material and the pattern region comprises a second dielectric material.

Example 29 includes an integrated circuit (IC) package, comprising an IC die, including a base structure, a pattern region located on the base structure, wherein a recess is located between a first portion of the pattern region and a second portion of the pattern region, and wherein a sidewall of the first portion of the pattern region abuts the recess and a sidewall of the second portion of the pattern region abuts the recess, and a growth base region located on the base structure and within the recess, the growth base region comprising a metal and oxygen, a self-assembled monolayer (SAM) material, or an organic material, wherein less than an entirety of the sidewall of the first portion of the pattern region abuts the growth base region and less than an entirety of the sidewall of the second portion of the pattern region abuts the growth base region, and a further IC component coupled to the IC die.

Example 30 includes the IC package of example 29, wherein the IC die further includes a growth region located on the growth base region and within the recess, wherein the growth region and the growth base region fill the recess.

Example 31 includes the IC package of example 30, wherein the growth region abuts a portion of the sidewall of the first portion of the pattern region and a portion of the sidewall of the second portion of the pattern region.

Example 32 includes the IC package of example 31, wherein the portion of the sidewall of the first portion of the pattern region is a first portion of the sidewall of the first portion of the pattern region, wherein the portion of the sidewall of the second portion of the pattern region is a first portion of the sidewall of the second portion of the pattern region, and wherein the growth base region abuts a second portion of the sidewall of the first portion of the pattern region and a second portion of the sidewall of the second portion of the pattern region.

Example 33 includes the IC package of example 29, wherein the IC die further includes a growth region located on the growth base region and within the recess, and a protective region, a first portion of the protective region located between the growth region and the first portion of the pattern region and a second portion of the protective region located between the growth region and the second portion of the pattern region.

Example 34 includes the IC package of example 33, wherein the SAM material is a first SAM material, wherein the growth base region comprises the first SAM material or an organic material, and wherein the protective region comprises a second SAM material, the second SAM material being different from the first SAM material.

Example 35 includes the IC package of example 33, wherein the growth base region, the growth region, and the protective region fill the recess.

Example 36 includes a method of producing an integrated circuit (IC) structure, comprising applying a growth base material to a base assembly, the base assembly having a recess and the growth base material being applied to sidewalls and a bottom of the recess, directing etching ions at the growth base material to remove a portion of the growth base material, the etching ions directed at an angle of less than 90 degrees to a top surface of the growth base material, wherein the etching ions remove the growth base material from a portion of the sidewalls and the growth base material remaining on the base assembly forms a growth base region, and growing a growth region on the growth base region.

Example 37 includes the method of example 36, wherein the etching ions are directed at an angle of between 60 degrees and 85 degrees to the bottom of the recess.

Example 38 includes the method of example 36, wherein the base assembly comprises one or more dielectric materials, and wherein the growth base material comprises a metal and oxygen, a self-assembled monolayer (SAM) material, or an organic material.

Example 39 includes the method of example 38, wherein the SAM material is a first SAM material or an organic material, and wherein the method further comprises forming a protective region on the portion of the sidewalls, the protection region comprising a second SAM material.

Example 40 includes a method of producing an integrated circuit (IC) structure, comprising forming a first material on a base assembly, the base assembly having a recess and the first material being applied to sidewalls and a bottom of the recess, forming a sacrificial material on the first material and within the recess, wherein the first material is between the sacrificial material and the bottom of the recess, performing an etch of the first material to remove the first material from a portion of the sidewalls of the recess, wherein the sacrificial material prevents removal of the first material from a portion of the bottom of the recess, removing the sacrificial material from the first material, and forming a second material on the first material.

Example 41 includes the method of example 40, wherein performing the etch includes directing ions perpendicularly to the bottom of the recess at the first materials, the ions to remove the first material from the portion of the sidewalls of the recess and the sacrificial material prevents contact of the ions with the first material on the portion of the bottom of the recess.

Example 42 includes the method of example 40, wherein the base assembly comprises one or more dielectric materials, and wherein the first material comprises a metal and oxygen, a self-assembled monolayer (SAM) material, or an organic material.

Example 43 includes the method of example 40, wherein forming the second material includes filling the recess with the second material.

Example 44 provides an IC package that includes an IC die, the IC die including the IC structure according to any one of the preceding examples (e.g., any one of examples 1-43), and a further IC component, coupled to the IC die.

Example 45 provides the IC package according to example 44, where the IC die includes the IC structure according to any one of the preceding examples, e.g., the IC structure according to any one of examples 1-43.

Example 46 provides the IC package according to any one of examples 44-45, where the further IC component includes one of a package substrate, an interposer, or a further IC die.

Example 47 provides an electronic device that includes a carrier substrate; and an IC die coupled to the carrier substrate, where the IC die includes the IC structure according to any one of examples 1-43, and/or is included in the IC package according to any one of examples 45-46.

Example 48 provides the electronic device according to example 47, where the electronic device is a wearable or handheld electronic device.

Example 49 provides the electronic device according to examples 47 or 48, where the electronic device further includes one or more communication chips and an antenna.

Example 50 provides the electronic device according to any one of examples 47-49, where the carrier substrate is a motherboard.

Example 51 provides the electronic device according to any one of examples 47-50, where the electronic device is an RF transceiver, an RF receiver, or an RF transmitter.

Example 52 provides the electronic device according to any one of examples 47-50, where the electronic device is one of a switch, a power amplifier, a low-noise amplifier, a filter, a filter bank, a duplexer, an upconverter, or a downconverter of an RF communications device, e.g., of one of an RF transceiver, an RF receiver, or an RF transmitter.

The above description of illustrated implementations of the disclosure, including what is described in the Abstract, is not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. While specific implementations of, and examples for, the disclosure are described herein for illustrative purposes, various equivalent modifications are possible within the scope of the disclosure, as those skilled in the relevant art will recognize. These modifications may be made to the disclosure in light of the above detailed description.

The invention claimed is:

1. An integrated circuit (IC) structure, comprising: a first dielectric material; a second dielectric material on the first dielectric material; a recess in the second dielectric material, wherein the recess includes a bottom, a top, a first sidewall, and a second sidewall, the bottom extending across a width of a recess from the first sidewall to the second sidewall, wherein a top surface of the first dielectric material defines the bottom of the recess, the second dielectric material extends along the heights of the first sidewall and the second sidewall, and the bottom of the recess is aligned with a bottom surface of the second dielectric material; a first material within the recess and at the bottom of the recess, wherein the first material includes a metal and oxygen, a self-assembled monolayer (SAM) material, or an organic material, and wherein the first material is directly on the first dielectric material; a second material within the recess and between the first material and the top of the recess, wherein a base of the second material is in direct contact with the first material; and a third material within the recess, the third material different from the first material and different from the second material, wherein the third material is within; a first region along the first sidewall of the recess and in contact with an upper surface of the first material within the recess; and a second region along the second sidewall of the recess and in contact with the upper surface of the first material within the recess, wherein the second material is in contact with the first region and the second region, wherein the third material along the first and second sidewalls of the recess is in direct contact with the second material, the first and second sidewalls of the recess, and the upper surface of the first material within the recess.

2. The IC structure of claim **1**, wherein a top edge of the second material is aligned with the top of the recess.

3. The IC structure of claim **1**, wherein the second material comprises a dielectric material or a metal.

4. The IC structure of claim **1**, wherein the bottom of the recess abuts the first dielectric material, and wherein the first material is on the first dielectric material.

5. The IC structure of claim **1**, wherein the first material comprises silicon and oxide.

6. The IC structure of claim **1**, wherein the first material has an upper surface, a lower surface, and a side extending between the upper surface and the lower surface, the side at an acute angle relative to the lower surface.

7. The IC structure of claim **1**, wherein the first material has a first thickness proximate to the first sidewall and a second thickness at a midpoint between the first sidewall and the second sidewall, the first thickness greater than the second thickness.

8. The IC structure of claim **1**, wherein the first material includes the SAM material.

9. The IC structure of claim **1**, wherein the first material includes the organic material.

10. The IC structure of claim **1**, wherein the bottom of the recess is aligned with the top surface of the first dielectric material.

11. The IC structure of claim **1**, wherein the first material within the recess is under the first region comprising the third material and under the second region comprising the third material.

12. The IC structure of claim **1**, wherein the second region comprising the third material is in contact with a first portion of the second sidewall of the recess, and wherein the second material is in contact with a second portion of the second sidewall of the recess.

13. The IC structure of claim **1**, wherein, a width of the first material within the recess is greater than a width of the second material within the recess.

14. The IC structure of claim **1**, wherein a seam is not present in a cross-section through the second material.

15. The IC structure of claim **1**, wherein, during a deposition process, the second material grows over the first material, and the second material does not attach to the third material.

16. An integrated circuit (IC) package, comprising: an IC die, including: a first dielectric material; a second dielectric material on the first dielectric material, wherein a recess is between a first portion of the second dielectric material and a second portion of the second dielectric material, wherein a top surface of the first dielectric material defines a bottom of the recess, the bottom of the recess is aligned with a bottom surface of the second dielectric material, a first sidewall of the first portion of the second dielectric material abuts and extends up a height of the recess, and a second sidewall of the second portion of the second dielectric material abuts and extends up a height of the recess, wherein the bottom extends across a width of the recess from the first sidewall to the second sidewall; a first material directly on the first dielectric material and within the recess, the first material comprising a metal and oxygen, a self-assembled monolayer (SAM) material, or an organic material; a second material within the recess and between the first material and a top of the recess, wherein a base of the second material is in direct contact with the first material; and a third material within the recess, the third material different from the first material and different from the second material, wherein the third material is within: a first region along the first sidewall of the recess and in contact with an upper surface of the first material within the recess; and a second region along the second sidewall of the recess and in contact with the upper surface of the first material within the recess, wherein the second material is in contact with the first region and the second region; and a further IC component coupled to the IC die, wherein the third material along the first and second sidewalls of the recess is in direct contact with the second material, the first and second sidewalls of the recess, and the upper surface of the first material within the recess.

17. The IC package of claim **16**, wherein a height of the first material is less than a height of the recess.

18. The IC package of claim **16**, wherein the second material abuts the first region comprising the third material and the second region comprising the third material.

19. The IC package of claim **16**, wherein the SAM material is a first SAM material, wherein the first material comprises the first SAM material or an organic material, and wherein the third material comprises a second SAM material, the second SAM material being different from the first SAM material.

20. The IC package of claim **16**, wherein the first material, the second material, and the third material fill the recess.

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